T follicular helper cell profiles differ by malaria antigen and for children compared to adults

Reviewed Preprint
v2 • April 11, 2025
Revised by authors

Reviewed Preprint
v1 • June 21, 2024

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eLife Assessment

This descriptive study used multiparameter spectral flow cytometry and clustering analysis of a subset of CD4 T cells, termed circulating T follicular helper (cTfh), responding to Plasmodium falciparum antigens, PfSEA -1A and PfGARP. The results from this comprehensive study provide **valuable** information regarding differences in cTfh response profiles between children and adults living in malaria-endemic Kenya and thus offer a potential usefulness towards improving choices of antigen candidates for malaria vaccines. However, the analysis and interpretation of antigen-specific CD4 cTfh responses remain **incomplete**.

https://doi.org/10.7554/eLife.98462.2.sa4

Abstract

Background

Circulating T-follicular helper (cT_{FH}) cells have the potential to provide an additional correlate of protection against *Plasmodium falciparum* (*Pf)* as they are essential to promote B-cell production of long-lasting antibodies. Assessing the specificity of cT_{FH} subsets to individual malaria antigens is vital to understanding the variation observed in antibody responses and identifying promising malaria vaccine candidates.

Methods

Using spectral flow cytometry and unbiased clustering analysis, we assessed antigen-specific cT_{FH} cell recall responses *in vitro* to malaria vaccine candidates *Pf*-schizont egress antigen-1 (*Pf*SEA-1A) and *Pf*-glutamic acid-rich protein (*Pf*GARP) within a cross-section of children and adults living in a malaria-holoendemic region of western Kenya.

Findings

In children, a broad array of cT_{FH} subsets (defined by cytokine and transcription factor expression) were reactive to both malaria antigens, *Pf*SEA-1A and *Pf*GARP, while adults had a narrow profile centering on cT_{FH}17- and cT_{FH}1/17-like subsets following stimulation with *Pf*GARP only.

Interpretation

Because T_{FH}17 cells are involved in the maintenance of memory antibody responses within the context of parasitic infections, our results suggest that *Pf*GARP might generate longer-lived antibody responses compared to *Pf*SEA-1A. These findings have intriguing implications for evaluating malaria vaccine candidates as they highlight the importance of including cT_{FH} profiles when assessing interdependent correlates of protective immunity.

Introduction

Despite progress made to reduce the global malaria burden, *Plasmodium falciparum* (*Pf*) remains one of the leading causes of mortality among children under 5 years of age.[1] Unfortunately, progress has been impeded by a plateau in malaria control since 2015. Anti-malarial drug resistance,[2,3] and unprecedented logistical challenges during the COVID-19 pandemic that dramatically impacted the distribution of insecticide-impregnated mosquito nets led to an increase in malaria cases in children under 5 years of age from 17.6 million in 2015 to 19.2 million in 2021 in East and Southern African countries,[1,4] thus, contributing to reinvigorated prioritization of malaria vaccine initiatives.

After 30 years of development, the first malaria vaccine—RTS,S/AS01E—was approved by the World Health Organization in 2021 for use in children residing in malaria-endemic regions. However, the limited efficacy of RTS,S,[5] particularly against severe malaria[5], has motivated the search for additional vaccine candidate antigens, including blood-stage *Pf*-schizont egress antigen-1 (*Pf*SEA-1A)[6] and *Pf*-glutamic acid-rich protein (*Pf*GARP).[7] Antibodies against *Pf*SEA-1A correlate with significantly lower parasite densities in Kenyan adults and adolescents and substantially reduce schizont replication *in vitro*.[6,8] Whereas *Pf*GARP-specific antibodies kill trophozoite-infected erythrocytes in culture and confer partial protection against *Pf*-challenge in vaccinated non-human primates.[7] The presence of antibodies against *Pf*GARP was associated with a 2-fold-lower parasite density in Kenyan adults and adolescents.[7] Due to the complexity of the parasites' life cycle, there is consensus that a multi-valent vaccine would be more efficacious[9]. However, determining which vaccine candidate antigens hold the most promise for inclusion in next-generation malaria vaccines remains a challenge.

Many pathogens and vaccines engender protective antibody responses after a single or few exposures[10], characterized by the production of long-lived plasma cells and memory B cells.[11] Affinity maturation takes place in the germinal center (GC), where antigen-specific $CD4^{pos}$ T-follicular helper (T_{FH}) cells are required not only to provide cellular (CD40L) and molecular (IL-21, IL-4, and IL-13) signals to trigger B-cell proliferation but also to promote GC maintenance and plasmablast differentiation. The GC is also where higher affinity B cells outcompete B cells with lower affinity for T_{FH} help.[12,13] T_{FH} cells are defined by the expression of a combination of markers, starting with chemokine receptor CXCR5, which directs $CD4^{pos}$ T cells from the T-cell zone to engage with follicular B cells.[14,15] Once antigen-experienced T_{FH} cells leave the GC, they become circulating T_{FH} (cT_{FH}) cells and correlate with the generation of long-lasting antibody responses. Interestingly, cT_{FH} cells can also come from peripheral cT_{FH} precursor CCR7^{low}PD1^{high}CXCR5^{pos} cells; thus, they also have a GC-independent origin[16]. Expression profiles of CCR6 and CXCR3 categorize cT_{FH} subsets as follows: $cT_{FH}1$ (CCR6^{neg}CXCR3^{pos}), $cT_{FH}2$ (CCR6^{neg}CXCR3^{neg}), and $cT_{FH}17$ (CCR6^{pos}CXCR3^{neg})[17], whereas the expression of PD-1, ICOS, CD127, and CCR7 define their functional status: quiescent/central memory cT_{FH} (CCR7^{high}PD-1^{neg}ICOS^{neg}CD127^{pos}) or activated/effector memory cT_{FH} (CCR7^{low}PD-1^{pos}ICOS^{pos}CD127$^{low/neg}$).[17–19] In addition, transcription factors (i.e., Bcl6 and cMAF) and cytokines (i.e., interferon-gamma [IFNγ], IL-4, and IL-21) are crucial to further characterizing the role of each cT_{FH} subset within the context of their interactions with B cells to promote antibody production.[20–23] Bcl6 facilitates the production of IL-21 by T cells which aids B-cell affinity maturation and antibody production.[24–26] T_{FH} cells also secrete cytokines that align with their subset classifications (but are not limited to them), such as IFNγ ($T_{FH}1$), IL-4/IL-13 ($T_{FH}2$), IL-4/IL-5/IL-13 ($T_{FH}13$), and IL-17 ($T_{FH}17$), that direct antibody isotype class-switching and mediate effector functions.[12,18,20–22,27]

Several studies have characterized cT_{FH} subsets within the context of both adult and childhood *Pf*-malaria infections, as well as in healthy malaria-naïve volunteers under controlled malaria infection conditions.[28] In Mali, where malaria is seasonal, Obeng-Adjei and colleagues showed that T_H1-polarized cT_{FH} PD1^{pos}CXCR5^{pos}CXCR3^{pos} cells were preferentially activated in children and less efficient than CXCR3^{neg} cT_{FH} cells in helping autologous B cells produce antibodies; yet they used U.S. healthy adult cT_{FH} cells for their *in vitro* assays. Therefore, the authors suggested that promoting T_H2-like CXCR3^{neg} cT_{FH} subsets could improve antimalarial vaccine efficacy.[29] Similarly, a recent study in Papua, Indonesia, where malaria is perennial, found that all cT_{FH} subsets from adults showed higher activation and proliferation compared to cT_{FH} cells from children.[30] After *in vitro* stimulation with infected red blood cells, they found that all cT_{FH} subsets were activated in adults accompanied by IL-4 production, whereas only the T_H1-polarized cT_{FH} subset responded in children.[30] One study in Uganda, where malaria is holoendemic with seasonal peaks, showed that a shift from T_H2- to T_H1-polarized cT_{FH} subsets occurred during the first 6 years of life and was associated with the development of functional antibodies against *Pf*-malaria, yet appeared to be independent of malaria exposure as this age-associated shift was also observed in a malaria-naïve population.[31] Interestingly, this study also found that a higher proportion of T_H17-cT_{FH} cells was associated with a decreased risk of *P. falciparum* infection the following year; however, the authors postulated that this phenomenon could have been driven by previous exposure to the parasite. The observed higher abundance of T_H2-like cT_{FH} cells in children younger than 6 years old and the age-associated increase in T_H1-cT_{FH} cells achieving the same proportion of T_H1-cT_{FH} cells by 6 years of age and into adulthood appeared to be independent of malaria exposure. Of note, these studies were limited by the use of a 2-dimensional gating strategy to classify cT_{FH} subsets and whole parasite activation conditions, either during infections or *in vitro* stimulation assays, leaving the malaria-antigen specificity of the different cT_{FH} subsets responses undefined. However, these studies highlight the progression in the field of evaluating the role of human T_{FH} cells in anti-malarial immunity.

Antibody levels are an unreliable predictor of malaria vaccine efficacy.[32] Thus, establishing a combined immune profile to incorporate other surrogates of protection is warranted. Here, we examined the profile of cT_{FH} subsets using multiparameter spectral flow cytometry[33,34] against two malaria vaccine candidate antigens (*Pf*SEA-1A[6] and *Pf*GARP[7]) in a cross-section of children and adults residing in a malaria-holoendemic region of Kenya. Our findings revealed significant differences in cT_{FH} subsets between children and adults, where children had more abundant $cT_{FH}1$-like cells and showed antigen-specific responses from all cT_{FH} types, whereas these responses were limited to $cT_{FH}17$- and 1/17-like cells in adults. Moreover, this study showed that *Pf*GARP triggered Bcl6 expression across cT_{FH} subsets, whereas *Pf*SEA-1A induced more cMAF expression, demonstrating the feasibility of implementing T-cell immune correlates to down-select new malaria candidates.

Results

Children had lower anti-PfSEA-1A antibodies compared to adults but similar levels of anti-PfGARP antibodies

This cross-sectional study selected a convenience sample of 7-year-old children and adults with a mean age of 22.67 years (ranging from 19 to 30 years old). No statistical difference was observed (p=0.35 after Welch's test) regarding the absolute lymphocyte count (ALC) between children (mean of 373.1; standard deviation [SD] of 118.1) and adults (mean of 335.2; SD of 99.08). Children had significantly lower hematocrit values (median of 37%, interquartile range [IQR] of 32.7–40.5) compared to adults (median of 44.4%, IQR of 40.5– 46), p=0.002 after a two-tailed Mann-Whitney t-test; however, these values were within normal ranges after adjusting for age.[35] Both males and females (sex assigned at birth) were enrolled, with 43% (6/14) and 53% (8/15) being female within the children and adult groups, respectively. Seroprofiles against a panel of commonly used malaria antigens were generated to confirm the history of previous malaria infections for the selected children (**Figure 1a**). All participants had high IgG levels against merozoite antigens, apical membrane antigen 1 (AMA-1) and merozoite surface protein (MSP1), confirming at least one malaria infection within their lifetime.[36,37] IgG antibodies against circumsporozoite protein (CSP) and CelTOS (liver-stage antigens) were characteristically lower than against blood-stage antigens yet were present in all study children. We observed a clear bimodal distribution in antibody levels against histidine-rich protein 2 (HRP2) with half of the children having "high-HRP2" versus "low-HRP2" IgG levels, possibly a reflection of recent malaria history since it has been suggested that HRP2-specific antibodies are short-lived and could serve as a surrogate for a recent infection.[38] We assessed serological profiles for two *Pf*-malaria antigens being considered as potential vaccine candidates,[6,7] *Pf*SEA-1A and *Pf*GARP (**Figure 1b**). We found that children had a significantly lower median level of IgG against *Pf*SEA-1A compared to adults (p<0.0001), whereas median levels to *Pf*GARP were similarly high for adults and children, yet with a broad range of reactivity.

The overall abundance of $CD4^{pos}CXCR5^{pos}$ cells is unaltered by in vitro antigen stimulation

To determine whether *in vitro* antigen stimulation with *Pf*SEA-1A or *Pf*GARP altered the abundance of total cT_{FH} cells, we compared cT_{FH} cells from adults and children using a FlowSOM unbiased clustering analysis and EMBEDSOM dimensional reduction based on common lineage markers assessed by spectral flow cytometry from 87,812 live lymphocytes from each sample: $CD8^{pos}$, $CD4^{pos}$, $CD4^{pos}CXCR5^{pos}$, and $CD4^{pos}CD25^{pos}$ (**Figure 2a**). As expected, after a short stimulation (6 hrs), overall abundances of cT_{FH} cells were similar across conditions for both adults (**Figure 2b**) and children (**Figure 2c**). This observation was confirmed by EdgeR statistical analysis (Supplemental Figure 2) and demonstrated that *in vitro* stimulation did not preferentially expand the T cell populations on which we based our subsequent analyses.

Figure 1

Pf-malaria IgG sero-profiles for children and adults.

(a) IgG antibody levels against AMA1, MSP1, HRP2, CelTos, and CSP for children (n=14). The bar plots indicate the mean with SD. (**b)** IgG antibody levels against *Pf*SEA-1A and *Pf*GARP comparing children (n=15) and adults (n=15). The bar plots indicate the mean with SD. Net MFI values are the antigen-specific MFI values minus the BSA background. Mann-Whitney tests were performed.

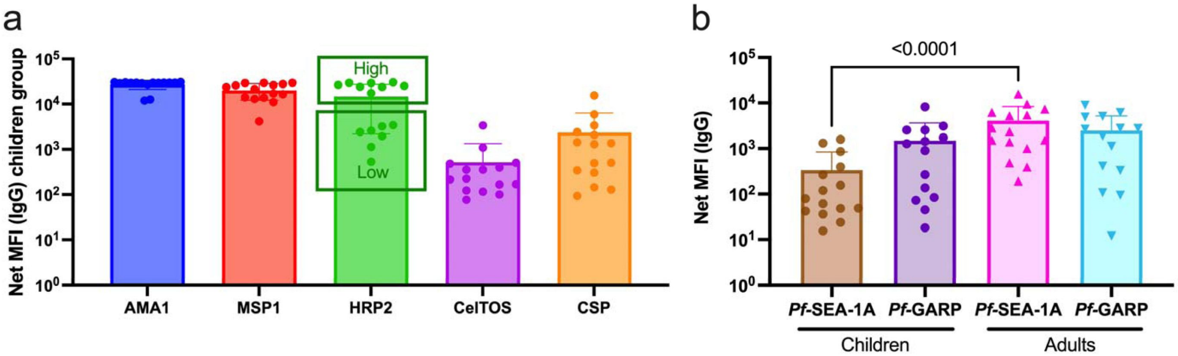
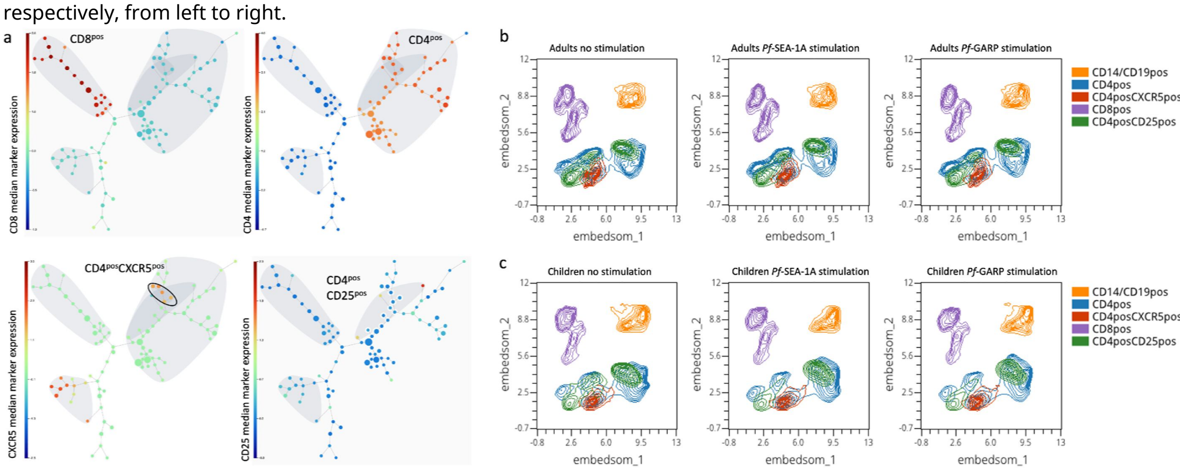


Figure 2

Cluster visualization of immune cell types and their abundance.

(a) A 100- node and 75-meta-cluster FlowSOM tree was generated on live lymphocytes from all our participants (n=29 children and adults), highlighting CD3^{pos}CD8^{pos}, CD3^{pos}CD4^{pos}, CD4^{pos}CXCR5^{pos}, and CD4^{pos}CD25^{pos} cells. The 5 nodes of the CD4^{pos}CXCR5^{pos}CD25^{neg} population are circled and were used for the downstream analysis. The colored scale was based on the median arcsinh-transformed marker expression. UMAP plots showing 5 clusters defined as follows: CD14^{pos} and CD19^{pos} (orange), CD4^{pos} (blue), CD4^{pos}CXCR5^{pos} (red), CD8^{pos} (purple), and CD4^{pos}CD25^{pos} (green) from PBMCs isolated from **(b)** adults (n=15) and **(c)** children (n=14) that were unstimulated, stimulated with *Pf*SEA-1A, or stimulated with *Pf*GARP, respectively, from left to right.



An unbiased clustering analysis identifies twelve distinct cT_{FH} meta-clusters

Numerous markers and a two-dimensional gating strategy have previously been used to determine the frequency of cT_{FH} subsets.[29–31] To simultaneously account for the expression of 17 markers required to define cT_{FH} subsets, we used FlowSOM unbiased clustering analysis to determine the frequency of cT_{FH} subsets from a pool of 1,000 $CD3^{pos}CD4^{pos}CXCR5^{pos}CD25^{neg}$ cells from each sample (total of 13,000 $CD3^{pos}CD4^{pos}CXCR5^{pos}CD25^{neg}$ cells in both children and adults). Based on CXCR3 and CCR6 expression as well as the expression of effector/memory/activation markers (CCR7, CD127, PD1, and ICOS), cytokines (IFNγ, IL-4, and IL-21), and transcription factors (Bcl6 and cMAF), we initially identified 15 meta-clusters within the $CD4^{pos}CXCR5^{pos}$ T cells (**Figure 3a**). First, we found different CXCR5 expression levels between meta-clusters (**Figure 3b**); CXCR5 is essential for cT_{FH} cells to migrate to the lymph nodes and interact with B-cells. Data presented in **Figure 3** are from children; however, we found similar observations in the adult group (Supplemental Figure 3). Because $CD45RA^{pos}CXCR5^{pos}$ cells are likely naïve cells with transient low expression of CXCR5 yet high expression of CD45RA, we excluded 3 meta-clusters using these criteria (i.e., MC12, MC14, and MC15) (**Figure 3c**). Then, using the overall expression of CXCR3 and CCR6 across the cT_{FH} subsets (**Figure 3d** and **3e**), we identified the remaining 12 clusters as follows: MC01 and MC02 were cT_{FH}2-like; MC06 and MC07 were cT_{FH}1-like; MC09 and MC11 were cT_{FH}1/17-like; MC10 and MC13 were cT_{FH}17-like. However, CXCR3 expression was not clearly delineated for some subsets and did not align with the conventional CCR6 vs. CXCR3 cytoplot (**Figure 3f** and **3g**, Supplemental Figure 4). Based on the heatmap (**Figure 3f**), MC03, MC04, MC05, and MC08 clusters appear closer to cT_{FH}2-like MC01 and MC02 clusters than cT_{FH}1-like clusters MC06 and MC07, suggesting that they might be part of the cT_{FH}2-like subset. But, based on their intensity of CXCR3 expression and their distribution across the CCR6 vs. CXCR3 cytoplot (Supplemental Figure 4), we defined MC03, MC04, MC05, and MC08 clusters as "undetermined" and would require additional cytokine and transcription factor analyses to fully categorize them.

Heterogeneity of activation/maturation markers within cT_{FH} subsets

By assessing the expression of CCR7, PD1, CD127, and ICOS (**Figure 4**) and following the three-dimensional expression patterns adapted from Schmitt et al.[17] (Supplemental Figure 5), we determined the activation state of each cT_{FH} meta-cluster and which markers created novel subsets. Data in **Figure 4** are from children; however, similar observations were made for adults (Supplemental Figure 6). As expected, none of the extracellular markers showed significant differences in expression patterns after a short 6-hour stimulation, thus, representing the cT_{FH} repertoire present within our study participants. Interestingly, the cT_{FH}1or2 subset (MC03) was the only meta-cluster with high expression of PD1 (**Figure 4a**) accompanied by high ICOS (**Figure 4b**), low CCR7 (**Figure 4c**), and low CD127 expression (**Figure 4d**), suggesting that MC03 was an activated/effector cT_{FH}1or2 subset. The MC04 subset had low CCR7 and high ICOS expression but low CD127 and intermediate PD1 expression, indicating that this cluster was a less activated/effector cT_{FH}1or2 cluster compared to MC03.

We also observed additional nuances in the expression pattern for other cT_{FH}1-like meta-clusters. The cT_{FH}1-like MC06 cluster had an activated/effector profile, whereas the cT_{FH}1-like MC07 cluster had a quiescent/effector profile. The cT_{FH}1or2-like MC05 and MC08 clusters were defined as quiescent/memory (overall high CCR7 expression) yet with very low expression of ICOS for the MC05 cluster.

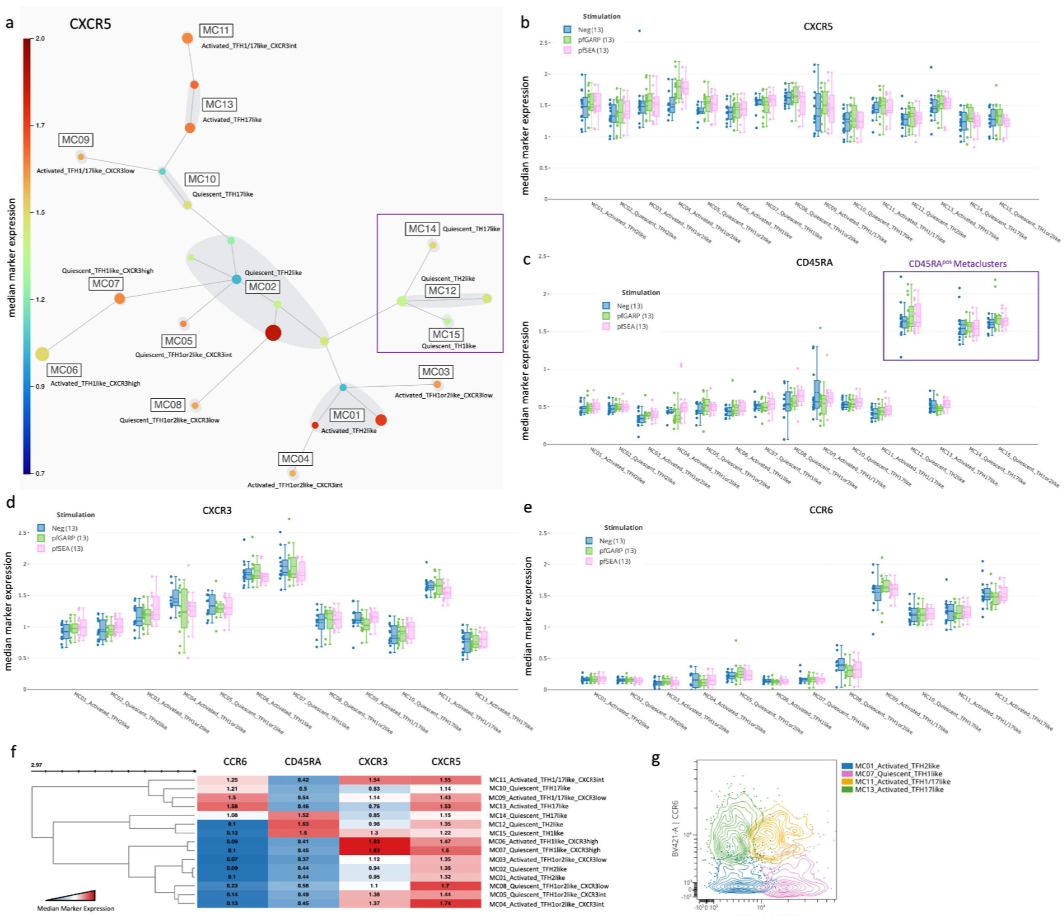


Figure 3

Characterization of T_{FH} subsets by CXCR5, CXCR3, CCR6, and CD45RA expression.

(a) A new 25-node and 15 meta-cluster FlowSOM tree was generated from the 5 CD4^{pos}CXCR5^{pos}CD25^{neg} nodes shown in **Figure 2A** (CXCR5 expression tree). The colored scale of CXCR5 expression was based on the median arcsinh-transformed with red as the highest and deep blue as the absence of CXCR5 expression. **(b)** Box plots showing CXCR5 expression across all cT$_{FH}$-like meta-clusters from children (n=13) with no stimulation (blue), and after *in vitro* stimulation with *Pf*GARP (green) or *Pf*SEA-1A (pink). Fifty percent of the data points are within the box limits, the solid line indicates the median, the dashed line indicates the mean, and the whiskers indicate the range of the remaining data with outliers being outside that range. Similar box plots are shown for **(c)** CD45RA, **(d)** CXCR3, and **(e)** CCR6 expression across all cT$_{FH}$-like meta-clusters. **(f)** Clustered heatmap showing the median arcsinh-transformed expression for CCR6, CD45RA, CXCR3, and CXCR5 across meta-clusters, red showing the highest expression and blue the lowest. **(g)** Cytoplots of CCR6 vs. CXCR3 expression where MC01 is blue, MC07 is pink, MC11 is yellow, and MC13 is green.

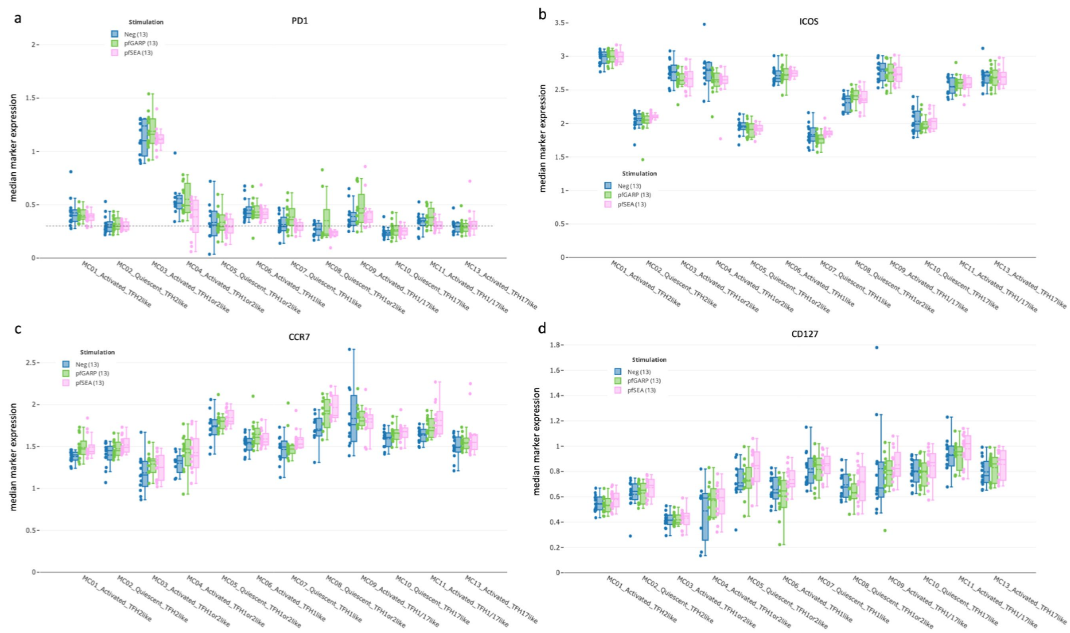


Figure 4

cT$_{FH}$ subset activation state determined by PD1, ICOS, CCR7, and CD127.

Box-plots showing **(a)** PD1, **(b)** ICOS, **(c)** CCR7, and **(d)** CD127 expression across all cT$_{FH}$- like meta-clusters from children (n=13) with no stimulation (blue), and after *in vitro* stimulation with *Pf*GARP (green) or *Pf*SEA-1A (pink). Fifty percent of the data points are within the box limits, the solid line indicates the median, the dashed line indicates the mean, and the whiskers indicate the range of the remaining data with outliers being outside that range.

Likewise, $cT_{FH}2$-like and $cT_{FH}17$-like meta-clusters also displayed heterogeneity. The $cT_{FH}2$-like MC01 cluster had an activated/effector profile, whereas the $cT_{FH}2$-like MC02 cluster seemed to be a quiescent/effector subset. The $cT_{FH}1/17$-like subsets MC09 and MC11 overall had an activated profile, although they had higher expression of CCR7 and CD127 compared to other subsets, suggesting an activated/memory phenotype. Finally, the $cT_{FH}17$-like MC10 cluster seemed quiescent, whereas $cT_{FH}17$-like MC13 had an activated/effector profile. Of note, because of our short stimulation time (6 hrs), we were unable to find statistical differences in the CD40L expression between groups as only few individuals responded through it (Supplemental Figure 7). However, our analysis methods revealed a higher degree of previously unrecognized heterogeneity within circulating cT_{FH} cells.

Activated $cT_{FH}1or2$-, $cT_{FH}1$-, and quiescent $cT_{FH}1or2$-like subsets were more abundant in children

After having deconvoluted cT_{FH} cells into 12 subsets, we next wanted to determine whether their abundance differed by age or after antigen stimulation. Using Uniform Manifold Approximation and Projection (UMAP) visualization, we found that cT_{FH} dimensional reduction was contiguous as meta-clusters merged with each other; in addition, there were notable differences between adults and children (**Figure 5a**). We found that activated $PD1^{high}$ $cT_{FH}1or2$-like (MC03), activated $cT_{FH}1$-like (MC06), and quiescent $ICOS^{high}$ $cT_{FH}1or2$-like (MC08) subsets were significantly more abundant in children compared to adults regardless of the stimulation conditions (**Figure 5b, 5c**, and **5d**, $p<0.05$). In contrast, the quiescent *Pf*SEA-1A- and *Pf*GARP-specific $cT_{FH}2$-like cluster (MC02) was significantly more abundant in adults compared to children (**Figure 5c** and **5d**, $p<0.05$). Interestingly, following *Pf*GARP stimulation, the activated $cT_{FH}1/17$-like subset (MC09) became more abundant in children compared to adults (**Figure 5d**, $p<0.05$ with a False Discovery Rate=0.08), but no additional subsets shifted phenotype after *Pf*SEA-1A stimulation (**Figure 5c**). Of note, the activated $PD1^{low}$ $cT_{FH}1or2$-like cells (MC04) seemed more abundant in non-stimulated adults and *Pf*GARP-stimulated children, but, because these observations were not present in all the participants, they did not achieve statistical significance in EdgeR.

The abundance heatmap (**Figure 5e**) reiterates the differences observed between children and adults and highlights important considerations when assessing the potential role of each cT_{FH} subset in assisting with cognate antibody production. Overall, the most common cT_{FH} subset in both children and adults is the quiescent $cT_{FH}2$-like cells (MC02, 24.1% and 41.2% respectively). However, the antigen-specific differences in the cT_{FH} subset abundance for children (MC09 for *Pf*GARP) and for adults (MC02 for both *Pf*SEA-1A and *Pf*GARP) suggest that children engage different cT_{FH} cells as they are developing immunity. Of note, only the activated $PD1^{low}$ $cT_{FH}1or2$-like cells (MC04) was less abundant in children with low compared to high HRP2 antibody levels (Supplemental Figure 8), suggesting that this subset may be involved in short-term antibody production. Overall, this comprehensive examination of the abundance of cT_{FH} subsets demonstrates important diversity based on age and malaria-antigen specificity.

PfSEA-1A and PfGARP induced IL-4, Bcl6, and cMAF from a broad range of cT_{FH} subsets in children

Because the children in this cohort were 7 years of age and resided in a malaria-holoendemic area, they had ample time to develop premunition. To assess antigen-specific cytokine and transcription factor expression signatures and further characterize cT_{FH} subsets, we generated clustered heatmaps of the median fluorescence intensity (MFI) of each analyte (IFN, IL-4, IL-21, Bcl6, and cMAF) for children (Supplemental Figure 9a) and adults (Supplemental Figure 9b). Next, using these MFI data, we performed Wilcoxon paired two-tailed t-tests to compare *Pf*SEA-1A and *Pf*GARP stimulation to unstimulated cells (**Figure 6a** and **6b**, respectively). Significant differences in these expression profiles allowed us to further characterize meta-clusters into three main groups. Group 1: $cT_{FH}2$-like (activated MC01 and quiescent MC02) and activated $cT_{FH}1or2$-like

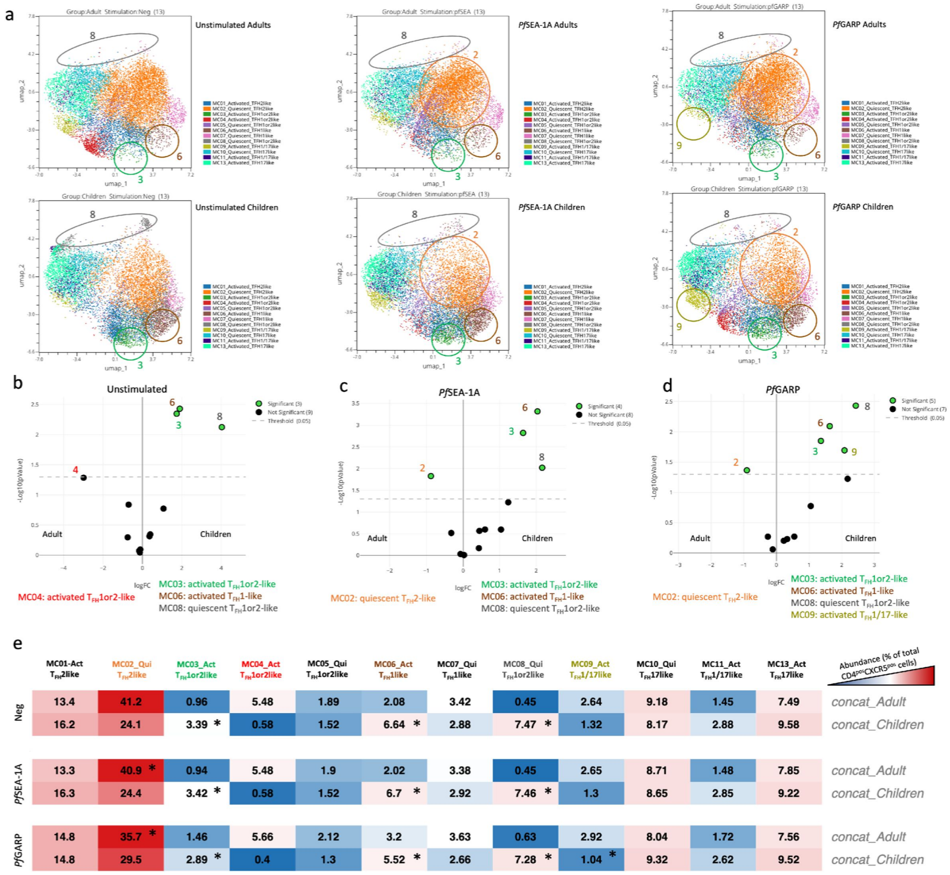


Figure 5

Differences in abundance of antigen-specific cT_{FH} meta-clusters for adults and children.

(a) UMAP plots showing the 12 different cT_{FH} meta-clusters in adults (top three plots, n=13) and in children (bottom three plots, n=13) in the absence of stimulation or after *in vitro* stimulation with *Pf*SEA-1A or *Pf*GARP, from left to right, respectively. Colored circles highlight the meta-clusters showing differences in their abundance between adults and children for each condition. An EdgeR statistical plot was performed to assess the change in abundance of the 12 meta-clusters between adults and children after **(b)** no stimulation or stimulation with **(c)** *Pf*SEA-1A or **(d)** *Pf*GARP. EdgeR plots indicate which meta-clusters are significantly abundant between two groups by using green color dots. The Y-axis is the - $\log_{10}$(*p*-value) and the X-axis is the log(FC). Green dots were statistically significant ($p<0.05$). Numbers next to the dots indicate the meta-cluster. **(e)** An abundance heatmap indicates the percentage (black numbers) of each meta-cluster within the total number of $CD3^{pos}CD4^{pos}CXCR5^{pos}CD25^{neg}$ cells for adults and children (concatenated from 13 participants in each group) under the different conditions: no stimulation (Neg) or stimulation with *Pf*SEA-1A or *Pf*GARP. The color scale ranges from high expression (red) to low/no expression (blue). The star in the heatmap indicates which meta-cluster is significantly abundant in children or adults based on the EdgeR results.

(PD1^{high}MC03 and PD1^{low}MC04); Group 2: activated and quiescent $cT_{FH}1$-like (MC06 and MC07); Group 3: activated $cT_{FH}1/17$-like (MC11) and $cT_{FH}17$-like (quiescent MC10 and activated MC13). For children, *Pf*SEA-1A and *Pf*GARP induced robust IL-4 expression in 9 out of 12 cT_{FH} meta-clusters (*p*-values ≤ 0.0105); although the composition of which cT_{FH} subsets were engaged differed slightly by antigen (quiescent $cT_{FH}1or2$-like ICOS^{low} MC05 versus quiescent $cT_{FH}2$-like MC02, respectively). In contrast to IL-4, we observed no change in expression for IFNγ or IL-21 after *in vitro* antigen stimulation (except for quiescent $cT_{FH}1or2$-like ICOS^{high} MC08, *p*- value=0.0391), suggesting that these cytokines are not informative to define the development of antigen-specific cT_{FH} subsets in children.

We found that *Pf*SEA-1A (**Figure 6a**) and *Pf*GARP (**Figure 6b**) induced similar Bcl6 and cMAF expression profiles from some of the same cT_{FH} subsets: both transcription factors were expressed by activated $cT_{FH}2$-like MC01 and quiescent $cT_{FH}17$-like MC10, but only cMAF was expressed in activated and quiescent $cT_{FH}1$-like MC06, MC07, and activated $cT_{FH}1/17$-like MC11 subsets. However, *Pf*SEA-1A induced Bcl6 and cMAF from activated PD1^{high}MC03, whereas *Pf*GARP only induced Bcl6. In contrast, the quiescent $cT_{FH}2$-like MC02 subset did not seem to respond to *Pf*SEA-1A (only cMAF was significant, *p*=0.0266, **Figure 6a**, Supplemental Figures 10 and 11), whereas *Pf*GARP stimulation induced significantly more IL-4, Bcl6, and cMAF compared to unstimulated cells (*p*=0.0081, *p*=0.0017, and *p*=0.0081, respectively, **Figure 6b**, Supplemental Figures 10 and 11). We then compared the response intensity between *Pf*SEA-1A and *Pf*GARP stimulations and found significant differences in IFNγ, IL-21, Bcl6, and cMAF expression levels (**Figure 6c**, 6d, 6e, and 6f, respectively). In Group 1, Bcl6 expression was significantly higher within activated and quiescent $cT_{FH}2$-like subsets (MC01 and MC02) as well as activated PD1^{high} $cT_{FH}1or2$-like (MC03) cells after *Pf*GARP compared to *Pf*SEA-1A stimulation (*p*=0.0266, *p*=0.0134 and *p*=0.0327, respectively, **Figure 6e**). Within Group 2, IFNγ and Bcl6 were highly expressed by activated $cT_{FH}1$-like (MC06) after *Pf*GARP stimulation compared to *Pf*SEA-1A stimulation (*p*=0.0093 and *p*=0.0024, respectively, **Figure 6c** and **6e**). Finally, within Group 3, the activated $cT_{FH}1/17$-like cells (MC11) expressed higher cMAF after *Pf*SEA-1A stimulation compared to *Pf*GARP (p=0.0186, **Figure 6f**). Interestingly, *Pf*GARP induced significantly more IL-21 within the quiescent ICOS^{low} $cT_{FH}1or2$-like (MC05) meta-cluster compared to *Pf*SEA-1A stimulation (*p*=0.0186, **Figure 6d**). Non-significant differences in cytokines and transcription factors expressed by cT_{FH} subsets between conditions are shown in Supplemental Figures 12 and 13, respectively.

PfGARP induced IL-4, Bcl6, and cMAF expression in activated $cT_{FH}1/17$- and $cT_{FH}17$-like subsets in adults

A similar heatmap was generated for adult expression profiles comparing *Pf*SEA-1A and *Pf*GARP stimulated to unstimulated cells (**Figure 7**). Here, we found that both *Pf*SEA-1A and *Pf*GARP induced significant expression of both IFNγ and IL-4 for activated (MC01) and quiescent (MC02) $cT_{FH}2$-like cells (**Figure 7a** and **7b**, Supplemental Figure 12). Whereas *Pf*GARP also induced IL-4 and IFNγ expression from quiescent and activated $cT_{FH}17$-like cells (MC10 and MC13), in addition to Bcl6 and cMAF for MC13 (**Figure 7b**, Supplemental Figure 13). This observation was surprising because IFNγ expression is commonly used to categorize the $cT_{FH}1$ subset (Group 2); however, as shown earlier, quiescent and activated $cT_{FH}17$-like cells (MC10 and MC13) did not express CXCR3 (**Figure 3d** and **3e**). In contrast, activated the $cT_{FH}1/17$-like cells (MC11) only responded to *Pf*GARP (**Figure 7b**, Supplemental Figure 12), expressing higher levels of IL-4, IL-21, Bcl6, and cMAF. Finally, while assessing the differences between the two malaria antigens, we found that *Pf*GARP induced more Bcl6 expression than *Pf*SEA-1A within the quiescent $cT_{FH}2$-like subset (MC02, *p*=0.0327) and the quiescent ICOS^{low} $cT_{FH}1or2$-like cells (MC05, *p*=0.0105), as well as within the quiescent $cT_{FH}17$-like subset (MC10, *p*=0.0479) and the activated $cT_{FH}1/17$-like cells (MC11, *p*=0.0020) (**Figure 7d**). The MC11 cells also expressed higher IL-21 levels after *Pf*GARP compared to *Pf*SEA-1A stimulation (*p*=0.0273, **Figure 7c**); whereas *Pf*SEA-1A induced cMAF within the activated $cT_{FH}2$-like subset (MC01, *p*=0.0134) but *Pf*GARP did not (**Figure 7e**). Overall, the main observation for adults is that *Pf*SEA-1A predominantly induced IL-4 from slightly more

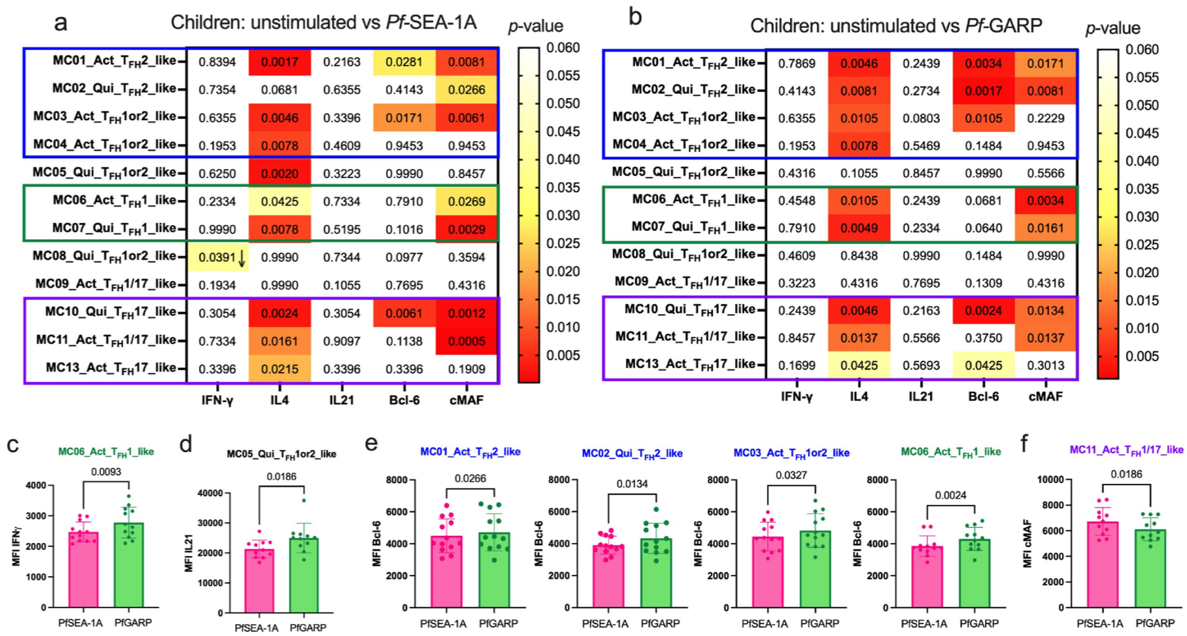


Figure 6

Diverse *Pf*-malaria antigen-specific expression patterns of cT_{FH}-defining cytokines and transcription factors for children.

Heatmaps of Wilcoxon-paired two-tailed t-test *p*-values are shown for cT_{FH} meta-clusters comparing **(a)** *Pf*SEA-1A and **(b)** *Pf*-GARP versus unstimulated PBMCs from children (n=13) for each cytokine (IFNγ, IL-4, and IL-21) and transcription factor (Bcl6 and cMAF). The color scale indicates the significance of the *p*- value: white (non-significant, $p>0.05$), yellow ($0.05>p>0.02$), orange ($0.02>p>0.005$), and red (highly significant, $p<0.005$). The down arrow indicates a decrease of expression from unstimulated to stimulated condition, whereas no arrow indicates an increase of expression from unstimulated to stimulated condition. The cT_{FH} meta-clusters that co-expressed transcription factors were grouped as follows: Group 1 (blue line), Group 2 (green line), and Group 3 (purple line). Bar plots indicating mean with SD of the median intensity fluorescence (MFI) of **(c)** IFNγ, **(d)** Bcl6, and **(e)** cMAF for the cT_{FH} meta-clusters showing significant statistical differences between *Pf*SEA-1A and *Pf*GARP stimulations. The *p*-values from Wilcoxon-paired two-tailed t-tests are indicated.

than half of the cT_{FH} clusters, whereas *Pf*GARP induced a broader range of cytokines and transcription factors but within the activated cT_{FH}1/17-like cells (MC11) and quiescent and activated cT_{FH}17-like subset (MC10 and MC13) similar to children. This analysis shows clear differences in cT_{FH} subset specificity by malaria antigen and cT_{FH} subset engagement by age group, with the cT_{FH} repertoire becoming more restricted in adults compared to children.

The activated cT_{FH}1or2-like subset is more abundant in participants with high anti-PfGARP antibodies

As shown in **Figure 1b**, a broad range of anti-*Pf*GARP IgG antibody levels were found in both children and adults. Thus, we wanted to determine whether the abundance of any of the cT_{FH} subsets was associated with the level of anti-*Pf*GARP IgG antibodies. When stratifying by high versus low anti-*Pf*GARP IgG antibody levels, we found that an activated cT_{FH}1or2-like subset (MC04) was more abundant in participants with high levels of anti-*Pf*GARP IgG for both children (**Figure 8a**) and adults (**Figure 8b**) after *Pf*GARP stimulation compared to participants with low levels or an absence of anti-*Pf*GARP IgG. However, even though the *p*- values were significant for both children and adults (p=0.02 and p=0.004, respectively), the FDR was less than 0.05 only for the adults (FDR=0.018). This suggests that this particular subset might be important for the generation of anti-*Pf*GARP antibodies. Interestingly, activated cT_{FH}1-like (MC06) and quiescent cT_{FH}1-like (MC07) cells were more abundant for adults with low or no anti-*Pf*GARP IgG antibodies compared to those with high levels (p=0.0024 with FDR=0.0185 and p=0.0034 with FDR=0.0185, respectively) consistent with previous observations describing T_{FH}1 subsets as inefficient help for antibody production[29].

Discussion

The overall aim of this study was to define cT_{FH} subsets using unbiased clustering analysis and assess their malaria antigen-specific (*Pf*SEA-1A and *Pf*GARP)[6,7,39] profiles for adults and children residing in a malaria-holoendemic area of Kenya. Contrary to previous publications,[29,30] our study found that children not only respond via their cT_{FH}1-like subsets but also engage a broader spectrum of cT_{FH} subsets. In fact, cytokine and transcription factor profiles for children involved cT_{FH}1-, cT_{FH}2-, cT_{FH}17-, and cT_{FH}1/17-like subsets (summarized in **Figure 9**), whereas in adults, the dominant antigen-specific memory response was from cT_{FH}17- and cT_{FH}1/17-like subsets but only for *Pf*GARP. Thus, revealing a potential difference in engaging cT_{FH} help between the two malaria vaccine candidates evaluated in this study.

Several key points can be made from our results. First, it may be too soon in the field of cT_{FH} biology to select only one or two types of cT_{FH} subset(s) to measure when evaluating malaria vaccine candidates, especially if we miss essential, albeit transient, correlates of protection for children which may differ in adults. Longitudinal studies with clinical outcomes are needed to determine the impact of the changing dynamics of cT_{FH} subset adaptations after repeated malaria infections that lead to premunition and parasite clearance. A Ugandan study by Chan et al.[31] demonstrated an age-associated change in cT_{FH} subsets independent of malaria and, thus, supports the importance of accounting for age when evaluating antigen-specific cT_{FH} profiles. Second, our study reveals that adults have a robust cT_{FH}17- and cT_{FH}1/17-like subset response to *Pf*GARP but not to *Pf*SEA-1A. In addition to demonstrating differences between malaria antigens, this finding supports the premise that cT_{FH}17 cells are important for maintaining immunological memory,[40] which has intriguing implications for evaluating malaria vaccine efficacy. Third, we found a correlation between cT_{FH}1or2-like (MC04) and high *Pf*GARP antibodies for both children and adults, indicating that a cT_{FH} subset accompanied by high antibody levels could serve as a potential biomarker of protection. More studies are needed to explore this association; however,

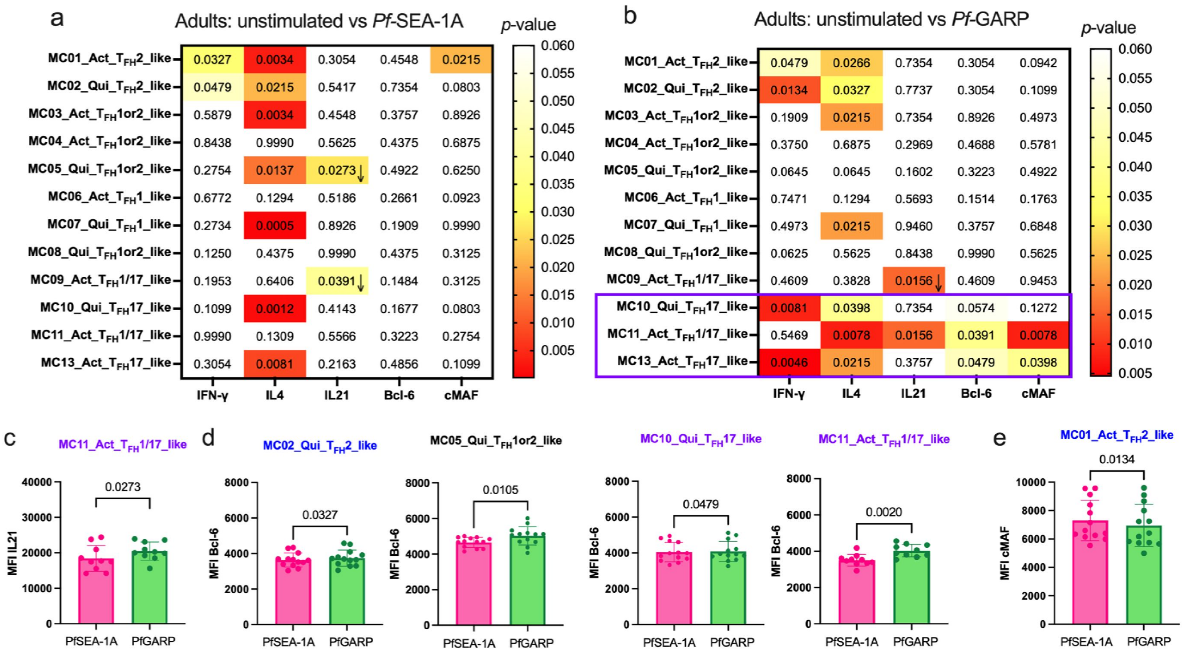


Figure 7

Limited *Pf*-malaria antigen-specific expression patterns of cT_{FH} defining cytokines and transcription factors for adults.

Heatmaps of Wilcoxon-paired two-tailed t- test *p*-values are shown for cT_{FH} meta-clusters comparing **(a)** *Pf*SEA-1A or **(b)** *Pf*-GARP versus unstimulated PBMCs from adults (n=13) for each cytokine (IFNγ, IL-4, and IL-21) and transcription factors (Bcl6 and cMAF). The color scale indicates the significance of the *p*-value: white (non-significant, p>0.05), yellow (0.05>p>0.02), orange (0.02>p>0.005), and red (highly significant, p<0.005). The down arrow indicates a decrease of expression from unstimulated to stimulated condition, whereas no arrow indicates an increase of expression from unstimulated to stimulated condition. The only cT_{FH} meta-clusters that expressed transcription factors were in Group 3 (purple box). Bar plots indicate the mean with SD of the median fluorescence intensity (MFI) of **(c)** Bcl6 and (**d**) cMAF for the cT_{FH} meta-clusters showing significant statistical differences between *Pf*SEA-1A and *Pf*GARP stimulations. The *p*-values from Wilcoxon-paired two-tailed t-tests are indicated.

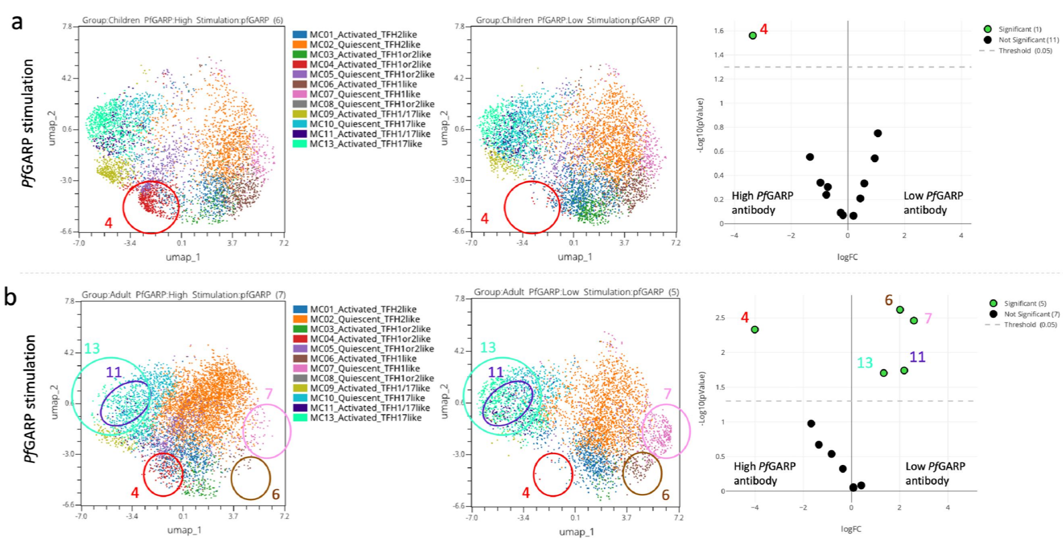


Figure 8

Abundance of cT_{FH} meta-clusters stratified by anti-*Pf*GARP IgG antibody levels.

UMAP and EdgeR analyses of **(a)** children (n=13) and **(b)** adults (n=13) where significant differences (p<0.05) in the abundance of the cT_{FH} meta-clusters are circled on high versus low *Pf*GARP antibody level (left to right) in the UMAP plot and green dots on the volcano plot (far right).

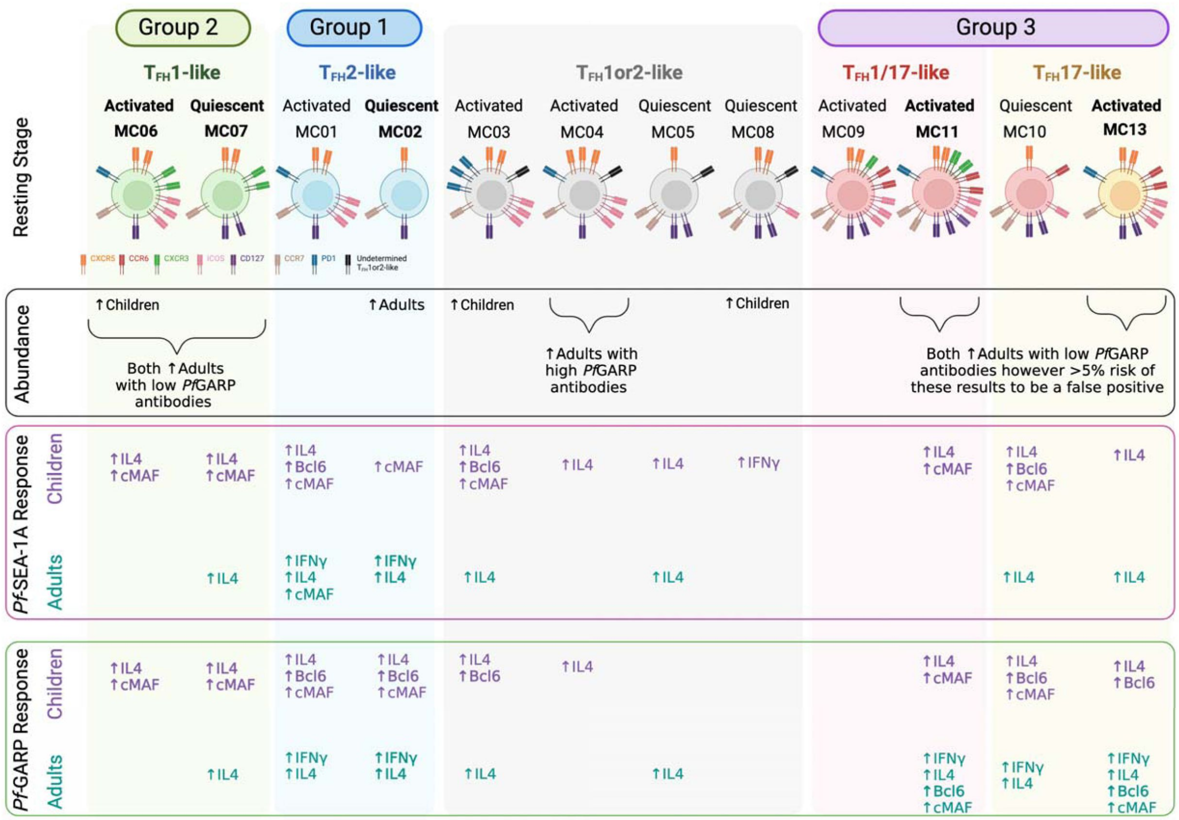


Figure 9

Summary.

Illustration of combined findings showing remarkable differences in cT_{FH} subset abundance and *Pf*-malaria antigen-specific cytokine and transcription factor responses between children and adults residing in a malaria-holoendemic region of Kenya. Figure created with BioRender.

we postulate that a coalition of cT_{FH} subsets might be engaged to develop long-lived antibody responses and, therefore, categorizing subsets as efficient or inefficient might be context dependent.[29]

As the field of computational immunology and unbiased clustering analyses evolves, it presents an ongoing challenge to meaningfully define cT_{FH} subsets. Here, we intentionally chose to use cT_{FH}1or2-like nomenclature for meta-clusters MC03, MC04, MC05, and MC08 (**Figure 9**) because of their low (but not null) CXCR3 and robust IL-4 expression along with Bcl6 and cMAF). CXCR3 expression leans toward a cT_{FH}1-like polarization, whereas IL-4 expression indicates a cT_{FH}2-like profile when we follow the most commonly used cT_{FH} classification.[17,21,22,33] This difference is crucial as cT_{FH}2 cells are described as good promoters for functional GC antibodies[28,30], whereas cT_{FH}1 cells are not.[29,30,41] Therefore, their definitive classification will require a more in-depth investigation. Although we used malaria antigen-specific stimulation, and not a *Pf*-lysate or infected red blood cells (iRBCs) as previously described,[29–31] adults had significantly more of the quiescent cT_{FH}2-like subset (MC02) compared to children who, instead, had significantly more of the activated cT_{FH}1- like subset (MC06), the latest being consistent with previous publications,[29,30] and more PD1^{high} activated and ICOS^{high} quiescent cT_{FH}1or2-like subsets (MC03 and MC08, respectively). However, even when classifying the cT_{FH} meta-clusters as cT_{FH}1, cT_{FH}2, cT_{FH}17, or cT_{FH}1/17, the most abundant cT_{FH} subset was cT_{FH}2 cells for adults (more than 50%) contradicting a previous study showing only ~20% of cT_{FH}2 cells for the same age range.[31] As suggested by Gowthaman's T_{FH} model,[27] T_{FH}1 cells are involved in the development of neutralization antibody responses to viruses and bacteria, whereas parasites, such as helminths, lead to a T_{FH}2 response. These observations reinforce the need to assess T_{FH} subset profiles stratified by exposure to potentially immune-modulating co-infections.

Transcription factors cMAF and Bcl6 play essential roles in T_{FH} development and function.[24,42,43] cMAF induces the expression of various molecules, such as ICOS, PD1, CXCR5, IL-4, and IL-21, which are all essential for T_{FH} function.[42] It was, therefore, not surprising to find that cMAF significantly increased after antigen stimulation for most of the cT_{FH} subsets in children. However, again, this expression profile was only observed in the activated cT_{FH}1/17-like subset (MC11) after *Pf*GARP stimulation in cells from adults, supporting a role for T_{FH}17 cells in the maintenance of immunological memory.[40] Importantly, cMAF cooperates with Bcl6 in T_{FH} development and function and is essential to establish an efficient GC response.[24,44] Bcl6 is also described as being expressed by mature T_{FH} cells, inducing the expression of CXCR5 and PD1, but it can also regulate IFNγ and IL-17 production.[12,24,44,45] In our study, *Pf*GARP induced a highly significant increase of Bcl6 in cT_{FH}2-like and cT_{FH}17-like subsets in children, suggesting that *Pf*GARP may be a better candidate to trigger an efficient humoral response in children compared to *Pf*SEA-1A. Of note, Blimp1, a Bcl6 antagonist,[45] was absent from our flow panel and would be of interest to assess in future studies.

There are several limitations of human immune profiling studies. Similar to other such studies, we used peripheral blood and, thus, were only able to provide a snapshot of the cT_{FH} cells. Study participants did not have blood-stage malaria infections at the time of blood collection, and children were 7 years of age; thus, our profiles were by design meant to reflect cT_{FH} memory recall responses and demonstrate differences between children and adults. As this was not a birth cohort study design, the number of cumulative malaria infections could have confounded the association between malaria and cT_{FH} subsets, causing a certain cT_{FH} meta-cluster to arise. Most human immunology studies are unable to assess tissue-resident T_{FH} or T_{FH} in the lymph nodes. Therefore, we can only speculate on the composition of T_{FH} cells observed in children that were perhaps short-lived helper cells and not maintained as T cell memory cells or may have trafficked from the blood into tissues over time.[46] Using an *in vivo* non-human primate model, Potter et al., showed an entry rate of lymphocyte subsets into peripheral lymph nodes per hour of 1.54% and

2.17% for CD4^{pos} central memory (CD45^{neg}CCR7^{pos}) and effector memory (CD45^{neg}CCR7^{neg}) T cells, respectively,[46] both of which may include central and effector memory cT$_{FH}$ cells. With this in mind, cT$_{FH}$ cells seem to be at a crossroads with multiple possible fates. The first one being the recirculation of cT$_{FH}$ cells from lymph node to lymph node, whereby a peripheral blood sampling captures migratory cT$_{FH}$ subsets, which could explain the cT$_{FH}$ subsets we observed still expressing cMAF and Bcl6. A decade ago, it was hypothesized that once a T$_{FH}$ cell leaves the lymph node, it can become a PD1^{low} memory cT$_{FH}$ cell with the possibility of returning to a GC T$_{FH}$ stage after a secondary recall response.[12] Other possible fates include becoming a PD1^{neg} memory cT$_{FH}$ with the same path after recall or progressing to a non-T$_{FH}$ cell.[12] More recently, a few studies showed circulating CXCR5^{neg}CD4^{pos} cells with T$_{FH}$ functions, such as production of IL-21 and the capability to help B cells in systemic lupus erythematosus and HIV-infected individuals,[47,48] and importantly researchers mapped these circulating CXCR5^{neg}CD4^{pos} cells to an original lymph node CXCR5^{pos} T$_{FH}$ subset,[48] suggesting another possible outcome for T$_{FH}$ cells outside secondary lymphoid organs. To date, no clear destiny has been established for cT$_{FH}$ cells in humans, thus, highlighting the importance of including a complete panel of cT$_{FH}$ subsets to continue to improve our understanding of their respective roles against different pathogens and eliciting long-lived vaccine-induced antibody responses.

Other limitations of this study include not measuring other cytokines (i.e., IL-5, IL-13, and IL-17) and transcription factors (i.e., T-bet, BATF, GATA3, and RORγt) that have been used in other studies to fully characterize cT$_{FH}$ subsets[12,27]. Although the heterogeneity in the response of CD40L and IFNγ suggests that our tested malaria antigens did not induce significant differences in the expression of these markers in all our participants, our panel did not include other activated induced markers, such as OX40, 4-1BB, and CD69. However, even with our small sample size, we demonstrated significant age-associated differences in malaria antigen-specific responses from different cT$_{FH}$ subsets. To minimize false-positive results that can arise when using algorithms for computational analyses, we ran the statistical tests in triplicate and the clustering algorithm in duplicate, to validate our findings.

In summary, our study provides additional justification for the resources needed to conduct cellular immunological studies of cT$_{FH}$ cell signatures and provides insight into which types of cT$_{FH}$ subsets during a person's lifespan assist B cells in the GC to produce long-lived plasmocytes and functional antibodies[49] against malaria. This is particularly important when selecting immune correlates of protection that could be used to predict the efficacy of the next generation of *Pf*-malaria vaccine candidates within various study populations.

Methods

Study populations and ethical approvals

Adults and children were recruited from Kisumu County, Kenya, which is holoendemic for *Pf*-malaria. Written informed consent was obtained from each adult participant and every child's guardian. An abbreviated medical history, physical examination, and blood film were used to ascertain health and malaria infection status at the time of blood sample collection. Participants were also life-long residents of the study area with an assumption that they naturally acquired immunity to malaria. This study was conducted before the implementation of any malaria vaccines. Participants were eligible if they were healthy and not experiencing any symptoms of malaria at the time venous blood was collected. For this cross-sectional immunology study, we selected fourteen 7-year-old children from a larger age-structured prospective cohort study (enrollment age range 3–7 years) and fifteen Kenyan adults. We selected 7-year-olds because of the age-dependent shift in major cT$_{FH}$ subsets occurring after 6 years of age[31], and as a comparable age published by other studies[29], to maximize our ability to measure antigen-specific differences in T$_{FH}$ subsets.

Ethical approvals were obtained from the Scientific and Ethics Review Unit (SERU) at the Kenya Medical Research Institute (KEMRI) reference number 3542, and the Institutional Review Board at the University of Massachusetts Chan Medical School, Worcester, MA, USA, IRB number H00014522. Brown University, Providence, RI, USA signed a reliance agreement with KEMRI.

Plasma and peripheral blood mononuclear cell (PBMC) isolation

Venous blood was collected in sodium heparin BD Vacutainers and processed within 2 hours at the Center for Global Health Research, KEMRI, Kisumu. Absolute lymphocyte counts (ALC) were determined from whole blood using BC-3000 Plus Auto Hematology Analyzer, 19 parameters (Shenzhen Mindray Bio-Medical Electronics Co.). After 10 mins at 1,000*g* spin, plasma was removed and stored at −20 °C, and an equivalent volume of 1X PBS was added to the cell pellet. PBMCs were then isolated using Ficoll-Hypaque density gradient centrifugation on SepMate (StemCell). PBMCs were frozen at 5×10^6 cells/ml in a freezing medium (90% heat-inactivated and filter-sterilized fetal bovine serum [FBS] and 10% dimethyl sulfoxide [DMSO, Sigma]) and chilled overnight in Mr. Frosty™ containers at −80 °C before being transferred to liquid nitrogen. For transport to the USA, an MVE vapor shipper (MVE Biological Solutions) was used to maintain the cold chain.

In-vitro stimulation assay

PBMCs were thawed in 37 °C filtered-complete media (10% FBS, 2 mM L-Glutamine, 10 mM HEPES, 1X Penicillin/Streptomycin) and spun twice before resting overnight in a 37 °C, 5% CO_2 incubator. PBMCs were counted using Trypan Blue (0.4%) and a hemocytometer, and the cell survival was calculated. Our samples showed a median of 94.6% live cells (25% percentile of 92%; 75% percentile of 97%). Using a P96 U-bottom plate, 1×10^6 PBMCs per well were placed in culture with one of the following stimulation conditions: *Pf*SEA-1A[6] (5 µg/ml) or *Pf*GARP[7] (10 µg/ml) both produced in the Kurtis lab (Brown University); SEB (1 µg/ml; EMD Millipore) was used as a positive control; sterile water (10 µl, the same volume used to reconstitute *Pf*SEA and *Pf*GARP) was used as a negative control. A pool of anti-CD28/anti-CD49d (BD Fast-Immune Co-Stim following the manufacturer's instructions), GolgiSTOP (0.7 µg/ml), and GolgiPLUG (0.1 µg/ml) (BD Biosciences) were added to each well before incubating cells at 37 °C for 6 hours.

Cell staining and flow cytometry

A multiparameter spectral flow cytometry panel was used to characterize cT_{FH} cell subsets: CCR6-BV421 (RRID: AB_2561356), CD14-Pacific Blue (RRID: AB_830689), CD19-Pacific Blue (RRID: AB_2073118), CCR7-BV480 (RRID: AB_2739502), IFNγ-BV510 (RRID: AB_2563883), CD127-BV570 (RRID: AB_10900064), CD45RA-BV605 (RRID: AB_2563814), PD1-BV650 (RRID: AB_2738746), CXCR3-BV711 (RRID: AB_2563533), CD25-BV750 (RRID: AB_2871896), CXCR5-BV785 (RRID: AB_2629528), Bcl6-AF488 (RRID: AB_10716202), CD3-Spark Blue 550 (RRID: AB_2819985), CD8-PerCP-Cy5.5 (RRID: AB_2044010), IL-21-PE (RRID: AB_2249025), IL-4-PE-Dazzle (RRID: AB_2564036), CD4-PE-Cy5 (RRID: AB_314078), ICOS-PE-Cy7 (RRID: AB_10643411), cMAF-eFluor 660 (RRID: AB_2574388), CD40L-AF700 (RRID: AB_2750053), and Zombie NIR (BioLegend cat# 423106) for Live/Dead staining. Cells were fixed and permeabilized for 45 mins using the transcription factor buffer set (BD Pharmingen) followed by a wash with the perm-wash buffer. Intracellular staining was performed at 4 °C for 45 more mins followed by two washes using the kit's perm-wash buffer. Data was acquired on a Cytek Aurora with 4-lasers (UMass Chan Flow Core Facility) using SpectroFlo® software (Cytek) and compensation for unmixing and fluorescence-minus-one controls. Quality control of the data was performed using SpectroFlo®, and the multi-parameter analysis was performed with OMIQ data analysis software (*www.omiq.ai*). Thus, we assessed the expression of markers commonly used to define the following different cT_{FH} ($CD4^{pos}CD25^{neg}CXCR5^{pos}$) subsets: cT_{FH}1-like ($CCR6^{neg}CXCR3^{pos}$), cT_{FH}2-like ($CCR6^{neg}CXCR3^{neg}$), and cT_{FH}17-like

(CCR6^{pos}CXCR3^{neg}),[17] as well as quiescent/central memory cT$_{FH}$ (CCR7^{high}PD-1^{neg}ICOS^{neg}) or activated/effector memory cT$_{FH}$ cells (CCR7^{low}PD-1^{pos}ICOS^{pos}).[17,18] Representative cytoplots can be found in Supplemental Figure 1.

Multiplex suspension bead-based serology assay

To measure plasma IgG antibody levels to *Pf*SEA-1A[6] and *Pf*GARP[7], we used a Luminex bead-based suspension assay as previously published.[50,51] In addition, previous *Pf* exposure was determined using recombinant proteins to blood-stage malaria antigens: AMA1, MSP1, HRP2, CelTos, and CSP (gifts from Sheetji Dutta, Evelina Angov, and Elke Bergmann from the Walter Reed Army Institute of Research). Briefly, 100 µg of each antigen or BSA (Sigma), as a background control, were coupled to ~12×10^6 non-magnetic microspheres (Bio-Rad carboxylated beads) and then incubated with study participant plasma (spun down 10,000*g* for 10 mins and diluted at 1:100 in the assay dilution buffer) for 2 hrs, followed by incubation with biotinylated anti-human IgG (BD #555785) diluted 1:1000 for 1 hr and streptavidin (BD #554061) diluted 1:1000 for 1 hr following the manufacturer's instructions. The mean fluorescence intensity (MFI) of each conjugated bead (minimum of 50 beads per antigen) was quantified on a FlexMap3D Luminex multianalyte analyzer (Xponent software). Results are reported as antigen-specific MFI after subtracting the BSA value for each individual since background levels can vary between individuals.

OMIQ analysis

The fcs files were uploaded into the OMIQ platform after passing quality control under SpectroFlo® (Cytek) where compensation was re-checked. In OMIQ, we arcsinh-transformed the scale to allow downstream analysis and then gated on singlet live lymphocytes and sub-sampled the data to yield 87,712 live lymphocytes per sample. Using only lineage markers CD3, CD4, CD8, CD14, CD19, CXCR5, and CD25, FlowSOM consensus meta-clustering was run on 100 clusters based on the 87,712 live lymphocytes per sample with a comma-separated k-value of 75 and Euclidean distance metric. Using these 75 meta-clusters, we defined subsets of cells based on lineage markers, such as CD3^{pos}CD8^{pos}, CD3^{neg}CD14^{pos}CD19^{pos}, and CD3^{pos}CD4^{pos}, and then distinguished CD4^{pos}CXCR5^{pos}CD25^{neg} (cT$_{FH}$) and CD4^{pos}CXCR5^{neg}CD25^{pos} (T regulatory [T$_{reg}$] or T follicular regulatory [T$_{FR}$]) subsets. CD25 was used to exclude T$_{reg}$ and T$_{FR}$ cells which share numerous markers with cT$_{FH}$ cells.[52–54] EmbedSOM dimensional reduction was used to visualize the different groups of cells and EdgeR analysis was run to assess the significance of their differences. A clustering heatmap was used to visualize cytokine expression and transcription factor profiles for each subset.

Focusing on the CD4^{pos}CXCR5^{pos}CD25^{neg} T$_{FH}$ cells, we ran another FlowSOM analysis based on the 1,000 CXCR5^{pos} cells per sample (two samples from the adult group and one sample from the children group were excluded from the analysis as they had less than 1,000 CXCR5^{pos} cells), using extracellular markers CXCR5, CXCR3, CCR6, ICOS, CCR7, CD45RA, CD127, CD40L, and PD1 enabled the identification of 15 meta-clusters. If the fcs file had more than 1,000 CXCR5^{pos} cells, the down-sampling was done randomly by the OMIQ platform algorithm to select only 1,000 CXCR5^{pos} cells within this specific fcs file. From there, we performed Uniform Manifold Approximation and Projection (UMAP) dimensional reduction, heatmaps, and EdgeR analyses; the latter allowed statistical analysis of the cT$_{FH}$ abundances. To demonstrate the reproducibility of these results, statistical analysis algorithms were run at least three times downstream of the same clustering algorithm and downstream of repeated clustering algorithms. To assess statistical differences in cytokines and transcription factor expression, we exported the statistics dataset from OMIQ containing MFI values from each marker (IFNγ, IL-4, IL-21, Bcl6, and cMAF) per cluster and for each sample and stimulation condition. To assess cytokines and transcription factors without bias, we chose to use the total MFI expression per meta-cluster with the assumption that cells with increased production of the desired analyte trigger an increase in the overall meta-cluster MFI compared to unstimulated cells, and if there is no production of the desired analyte, the overall

MFI will not differ. However, the percentage of positive IFNγ, IL-4, IL-21, Bcl6, or cMAF using manual gating can be found in Supplemental Figures 14, 15, and 16 along with the overlay of the gated positive cells on the $CD4^{pos}CXCR5^{pos}CD25^{neg}$ UMAP and the cytoplots of the gated positive cells for each meta-cluster (Supplemental Figures 14, 15, and 16).

Statistical analysis

For this cross-sectional immunology study, we selected both male and female study participants (sex defined at birth). There were fourteen 7-year-old children and fifteen adults. Using GraphPad Prism software (version 7.0), age, sex, ALC, and serological data were compared between adults and children. Because the number of participants within each group was too low to verify the normality of the underlying distributions (adults n=15 and children n=14), we chose to use non-parametric tests, including the Mann-Whitney U test (for unpaired analysis) and Wilcoxon signed-rank test (for paired analysis). When data passed the normality test (D'Agostino and Pearson test), we used Welch's parametric test. All tests were two-tailed with a *p-value* < 0.05 for significance. Because of the exploratory nature of the analysis, we did not use any adjustment of the *p*-value for multiple comparisons. The tests used are indicated in the legend of each figure. Results were expressed as the mean with standard deviation and exact *p*-values for dot plots.

To perform a statistical analysis of the cytokine and transcription factor expression from each cT_{FH} subset, the exported data file from OMIQ was integrated into GraphPad Prism, and a non-parametric Wilcoxon paired two-tailed *t*-test analysis was done (n=13 in each group). As this analysis generated numerous bar plots (all included in the Supplemental Figures 12 to 15), to better visualize the cytokines and transcription factor patterns, the *p*- values obtained from each analysis are presented using a non-clustering heatmap.

Role of the funding source

The funders were not involved in the study design, sample collection, data analysis, interpretation of the data, or the writing of the manuscript and its decision to be submitted for publication.

Data availability

Deidentified raw data from this manuscript are available from the ImmPort platform under the accession study number SDY2534.

Acknowledgements

The authors would like to thank the children and their families for participating in this study. We thank the field and lab teams for their work collecting data and processing blood samples. We also thank Dr. Melanie Trombly from UMass Chan for proofreading our manuscript. This manuscript was approved for publication by KEMRI.

This study was supported by NIH R01 AI127699 (Kurtis).

Additional information

Contributors

CN, JMO, JK, and AMM designed the research study; CF and SPT conducted experiments; CF, JK, and AMM analyzed data; CF, SPT, JK, and AMM contributed to experimental design; HWW, BO, JMO, JK, and AMM conducted and supervised the clinical study and the sampling; CF generated figures; HWW and BO verified underlying data; CF and AMM led manuscript preparation with feedback from all the authors. All authors have read and approved the final version of the manuscript.

Additional files

Supplemental figures 1 to 16

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Reviewer #1 (Public review):

Summary:

This study aims to understand the malaria antigen-specific cTfh profile of children and adults living in malaria holoendemic area. PBMC samples from children and adults were unstimulated or stimulated with PfSEA-1A or PfGARP in vitro for 6h and analysed by a cTfh-focused panel. Unsupervised clustering and analysis on cTfh was performed. The main conclusions are: A) the children cohort has a more diverse (cTfh1/2/17) recall responses compared to adults (mainly cTfh17) and, B) Pf-GARP stimulates better cTfh17 responses in adults, thus a promising vaccine candidate.

Strengths:

This study is, in general, well-designed and with excellent data analysis. The use of unsupervised clustering is a nice attempt to understand the heterogeneity of cTfh cells.

Weaknesses:

The authors have provided additional data in Supplementary Figures 14-16. However, I remain concerned about whether cTfh cells are truly responding to antigen stimulation. In Supplementary Figure 15A-F, the IFNg responses appear as expected, SEB elicits the strongest response, as it stimulates bulk T cells, and the staining is promising, showing a clear distinction between IFNg+ and IFNg- populations. However, in Supplementary Figure 15I-N, the IL-21 secretion assay is concerning. The FACS plots make it difficult to distinguish IL-21+

from IL-21- cells, raising concerns about the validity of this analysis. Additionally, in panel J, the responses to PfSEA-1A or PfGARP appear even greater than those to SEB stimulation. In PBMCs, only a small percentage of T cells should be specific to a particular antigen. How can the positive control (SEB) produce a weaker response than stimulation with a specific antigen? This suggests that the IL-21 secretion assay may not have worked, making the authors' interpretation unreliable.

I also have similar concerns about the IL-4 secretion in Sup Figure 16. First, the FACS plot shows that appear double-positive for IL-21 and IL-4, so it suggests the staining may be due to autofluorescence rather than true cytokine signals. Also in B-C the responses of SEB stimulation is generally weaker than stimulated by one antigen, further questioning the reliability of the IL-4 assay. In summary, I am not convinced that the in vitro antigen stimulation assay worked as intended. Consequently, the manuscript's claims regarding PfSEA-1A- and PfGARP-specific cTfh responses are not sufficiently supported by the presented data.

https://doi.org/10.7554/eLife.98462.2.sa3

Reviewer #3 (Public review):

Summary:

The goal of this study was to carry out an in-depth granular and unbiased phenotyping of peripheral blood circulating Tfh specific to two malaria vaccine candidates, PfSEA-1A and PfGARP, and correlate these with age (children vs adults) and protection from malaria (antibody titers against Plasmodium antigens.) Authors further attempted to identify any specific differences of the Tfh responses to these two distinct malaria antigens.

Strengths:

The authors had access to peripheral blood samples from children and adults living in a malaria-endemic region of Kenya. The authors studied these samples using in vitro restimulation in the presence of specific malaria antigens. Authors generated a very rich data set from these valuable samples using cutting-edge spectral flow cytometry and a 21-plex panel that included a variety of surface markers, cytokines and transcription factors.

Update following first revision (R1) of the manuscript:

The authors have made a great effort to comprehensively address comments raised by the reviewers. In particular, clearly showing expression of ICOS and Bcl6 on CXCR5+ cells greatly strengthens the case for defining these cells as Tfh-like circulatory lymphocytes (cTfh).

Weaknesses:

Update following first revision (R1) of the manuscript:

Unfortunately, my main concern remains. As it stands, the study is not really on antigen-specific T cells, but rather on the overall CD4 T cell compartment plus or minus antigenic stimulation. Although authors used an in vitro restimulation strategy with malaria antigens, they do not focus on cells de-novo expressing activation markers as a result of restimulation, neither they use tetramers to detect antigen-specific T cells. Moreover, their data shows that the number of CXCR5+ CD4 T cells de-novo expressing activation markers and/or cytokines as a result of their in vitro restimulation is negligible, even when using a prototypic superantigen (SEB).

Thus, no antigen-specific CXCR5+ CD4 T cells could be analysed with the data that the authors provide in this manuscript.

https://doi.org/10.7554/eLife.98462.2.sa2

Reviewer #4 (Public review):

Summary:

This manuscript is a descriptive study of circulating T follicular helper (cTfh) responses to PfSEA -1A or PfGARP (targets of new antimalaria vaccine candidates) in PBMCs from a convenience sample of children (7 yrs of age) and adults living in a malaria holo endemic Kenya using multiparameter flow cytometry and clustering analysis. This cell type promotes B cell production of long-lived antimalarial antibodies to provide protection against malaria. They find that children had a wider cTFH cytokine and TF profile cellular response in comparison to adults who responded to both antigens but had a narrower response profile.

Strengths:

Carefully done study, very detailed, nice summary model at the end of the paper. The revision provides requested clarification on a number of issues, including CD40L expression which was not differentially expressed between groups. They add additional data into the supplemental files, including IL4 and IL21 data by presenting the cytoplots.

Weaknesses:

To know the significance of these cTfh cells for long-term protection of malaria requires functional and transfer experiments in animal models which is outside the scope of this work.

https://doi.org/10.7554/eLife.98462.2.sa1

Author response:

The following is the authors' response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *This study aims to understand the malaria antigen-specific cTfh profile of children and adults living in a malaria holoendemic area. PBMC samples from children and adults were unstimulated or stimulated with PfSEA-1A or PfGARP in vitro for 6h and analysed by a cTfh-focused panel. Unsupervised clustering and analysis on cTfh were performed.*
>
> *The main conclusions are:*
>
> *(1) the cohort of children has more diverse (cTfh1/2/17) recall responses compared to the cohort of adults (mainly cTfh17) and*
>
> *(2) Pf-GARP stimulates better cTfh17 responses in adults, thus a promising vaccine candidate.*
>
> *Strengths:*
>
> *This study is in general well-designed and with excellent data analysis. The use of unsupervised clustering is a nice attempt to understand the heterogeneity of cTfh cells. Figure 9 is a beautiful summary of the findings.*

Weaknesses:

(1) Most of my concerns are related to using PfSEA-1A and PfGARP to analyse cTfh in vitro stimulation response. In vitro, stimulation on cTfh cells has been frequently used (e.g. Dan et al, PMID: 27342848), usually by antigen stimulation for 9h and analysed CD69/CD40L expression, or 18h and CD25/OX40. However, the authors use a different strategy that has not been validated to analyse in vitro stimulated cTfh. Also, they excluded CD25+ cells which might be activated cTfh. I am concerned about whether the conclusions based on these results are reliable.

It has been shown that cTfh cells can hardly produce cytokines by Dan et al. However, in this paper, the authors report the significant secretion of IL-4 and IFNg on some cTfh clusters after 6h stimulation. If the stimulation is antigen-specific through TCR, why cTfh1 cells upregulate IL-4 but not IFNg in Figure 6? I believe including the representative FACS plots of IL-4, IFNg, IL21 staining, and using %positive rather than MFI can make the conclusion more convincing. Similarly, the author should validate whether TCR stimulation under their system for 6h can induce robust BCL6/cMAF expression in cTfh cells. Moreover, there is no CD40L expression. Does this mean TCR stimulation mediated BCl6/cMAF upregulation and cytokine secretion precede CD40L expression?

In summary, I am particularly concerned about the method used to analyse PfSEA-1A and PfGARP-specific cTfh responses because it lacks proper validation. I am unsure if the conclusions related to PfSEA-1A/PfGARP-specific responses are reliable.

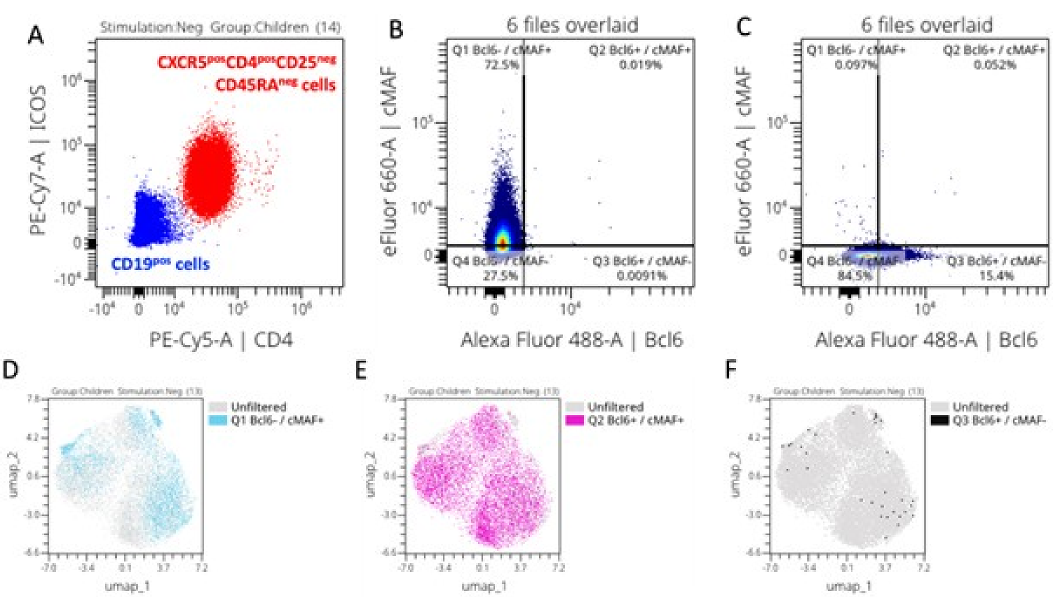
An unfortunate reality of these types of complex immunologic studies is that it takes time to optimize a multiparameter flow cytometry panel, run this number of samples, and then conduct the analysis (not to mention the time it takes for a manuscript to be accepted for peer-review). An unexpected delay, frankly, was the COVID-19 pandemic when non-essential research lab activities were put on hold. We designed our panel in 2019 and referred to the “T Follicular Helper Cells” Methods and Protocols book from Springer 2015. Obviously the field of human immunology took a huge leap forward during the pandemic as we sought to characterize components of protective immunity, and as a result there are several new markers we will choose for future studies of Tfh subsets. We agree with the reviewer that cytokine expression kinetics differ depending on the *in vitro* stimulation conditions. Due to small blood volumes obtained from healthy children, we were limited in the number of timepoints we could test. However, since we were most interested in IL21 expression, we found 6 hrs to be the best in combination with the other markers of interest during our optimization experiments. We did find IFNg expression from non-Tfh cells, therefore we believe our stimulation conditions worked.

Dan et al used stimulated tonsils cells to assess the CXCR5^{pos}PD1^{pos}CD45RA^{neg} Tfh and CXCR5^{neg} CD45RA^{neg} non-Tfh whereas in our study, we evaluated CXCR5^{pos}PD1^{pos}CD45RA^{neg} Tfh from PBMCs. Dan et al PBMCs’ work used EBV/CMV or other pathogen product stimuli and only gated on CD25^{pos}OX40^{pos} cells which are not the cells we are assessing in our study. This might explain in part the differences in cytokine kinetics, as we evaluated CD25^{neg} PBMCs only. However, we agree that more recent studies focused on CXCR5^{pos}PD1^{pos} cells included more Activation-induced marker (AIM) markers, which are missing in our study, inducing a lack of depth in our analysis.

Percentage of positive cells and MFI are complementary data. Indeed, the percentage of positive cells only indicates which cells express the marker of interest without giving a quantitative value of this expression. MFI indicates how much the marker of interest is expressed by cells which is important as it can indicate degree of activation or exhaustion per cell. Meta-cluster analysis is not ideal to assess the percentage of positivity whereas it does provide essential information regarding the intensity of expression. We added supplemental figures 14 (Bcl6 and cMAF), 15 (INFg and IL21) and 16 (IL4 and IL21) where percentage of positive cells were manually gated directly from the total CXCR5^{pos}CD4^{pos}CD45RA^{neg}CD25^{neg} TfH based on the FMO or negative control, and we overlaid the positive cells on the UMAP of all the CXCR5^{pos}CD4^{pos}CD45RA^{neg}CD25^{neg} meta-clusters. Results from the manual gating are consistent with the results we show using clustering. However, it helps to better visualize that antigen-specific IL21 expression was statistically significant in children whereas the high background observed for adults did not reveal higher expression after stimulation, perhaps suggesting an upper threshold of cytokine expression (supplemental figure 15). The following sentence has been added in the methods at the end of the "OMIQ analysis" section: " However, the percentage of positive IFNγ, IL-4, IL-21, Bcl6, or cMAF using manual gating can be found in Supplemental Figures 14, 15, and 16 along with the overlay of the gated positive cells on the CD4^{pos}CXCR5^{pos}CD25^{neg} UMAP and the cytoplots of the gated positive cells for each meta-cluster (Supplemental Figures 14, 15, and 16)."

Indeed cMAF can be induced by TCR signaling, ICOS and IL6 (Imbratta et. al, 2020). However, in our study populations, ICOS was expressed (see Author response image 1, panel A) in absence of any stimulation suggesting that CXCR5^{pos}CD4^{pos}CD25^{neg}CD45RA^{neg} cells were already capable of expressing cMAF. Indeed, after gating Bcl6 and cMAF positive cells based on their FMOs (Author response image 1, panel B and C, respectively), we overlaid positive cells on the CXCR5^{pos}CD4^{pos}CD25^{neg}CD45RA^{neg} cells UMAP and we can see that most of our cells already express cMAF alone (Author response image 1, panel D), co-express cMAF and Bcl6 (Author response image 1, panel E), confirming that they are TfH cells, whereas very few cells only expressed Bcl6 alone (Author response image 1, panel F). Because we knew that cT$_{FH}$ already expresses Bcl6 and cMAF, we focused our analysis on the intensity of their expression to assess if our vaccine candidates were inducing more expression of these transcription factors.

Author response image 1.



> *(2) The section between lines 246-269 is confusing. Line 249, comparing the abundance after antigen stimulation is improper because 6h stimulation (under Golgi stop) should not induce cell division. I think the major conclusions are contained in Figure 5e, that (A) antigen stimulation will not alter cell number in each cluster and (B) children have more MC03, 06 and fewer MC02, etc.). The authors should consider removing statements between lines 255-259 because the trends are the same regardless of stimulations.*

We agree, there is no cell division after 6h and that different meta clusters did not proliferate after this short of *in vitro* stimulation. The use of the word ‘abundance’ in the context of cluster analysis is in reference to comparing the contribution of events by each group to the concatenated data. After the meta clusters are defined and then deconvoluted by study group, certain meta clusters could be more abundant in one group compared to another - meaning they contributed more events to a particular metacluster.

Dimensionality reduction is more nuanced than manual gating and reveals a continuum of marker expression between the cell subsets, as there is no hard “straight line” threshold, as observed when using in 2D gating. Because of this, differences are revealed in marker expression levels after stimulation making them shift from one cluster to another - thereby changing their abundance.

To clarify how this type of analysis is interpreted, we have modified lines 255-259 as follows:

“In contrast, the quiescent *Pf*SEA-1A- and *Pf*GARP-specific $cT_{FH}2$-like cluster (MC02) was significantly more abundant in adults compared to children (Figure 5c and 5d, *pf*<0.05). Interestingly, following *Pf*GARP stimulation, the activated $cT_{FH}1/17$-like subset (MC09) became more abundant in children compared to adults (Figure 5d, *pf*<0.05 with a False Discovery Rate=0.08), but no additional subsets shifted phenotype after *Pf*SEA-1A stimulation (Figure 5c).”

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *Forconi et al explore the heterogeneity of circulating Tfh cell responses in children and adults from malaria-endemic Kenya, and further compare such differences following stimulation with two malaria antigens. In particular, the authors also raised an important consideration for the study of Tfh cells in general, which is the hidden diversity that may exist within the current 'standard' gating strategies for these cells. The utility of multiparametric flow cytometry as well as unbiased clustering analysis provides a potentially potent methodology for exploring this hidden depth. However, the current state of analysis presented does not aid the understanding of this heterogeneity. This main goal of the study could hopefully be achieved by putting all the parameters used in one context, before dissecting such differences into their specific clinical contexts.*
>
> *Strengths:*
>
> *Understanding the full heterogeneity of Tfh cells in the context of infection is an important topic of interest to the community. The study included clinical groupings such as age group differences and differences in response to different malaria antigens to further highlight context-dependent heterogeneity, which offers new knowledge to the field. However, improvements in data analyses and presentation strategies should be made in order to fully utilize the potential of this study.*
>
> *Weaknesses:*
>
> *In general, most studies using multiparameter analysis coupled with an unbiased grouping/clustering approach aim to describe differences between all the parameters used for defining groupings, prior to exploring differences between these groupings in specific contexts. However, the authors have opted to separate these into sections using "subset chemokine markers", "surface activation markers" and then "cytokine responses", yet nuances within all three of these major groups were taken into account when defining the various Tfh identities. Thus, it would make sense to show how all of these parameters are associated with one another within one specific context to first logically*

establish to the readers how can we better define Tfh heterogeneity. When presented this way, some of the identities such as those that are less clear such as "MC03/MC04/ MC05/ MC08" may even be better revealed. once established, all of these clusters can then be subsequently explored in further detail to understand cluster-specific differences in children vs adults, and in the various stimulation conditions. Since the authors also showed that many of the activation markers were not significantly altered post-stimulation thus there is no real obstacle for merging the entire dataset for the first part of this study which is to define Tfh heterogeneity in an unbiased manner regardless of age groups or stimulation conditions. Other studies using similar approaches such as Mathew et al 2020 (doi: 10.1126/science.abc8) or Orecchioni et al 2017 (doi: 10.1038/s41467-017-01015-3) can be referred to for more effective data presentation strategies.

Accordingly, the expression of cytokines and transcription factors can only be reliably detected following stimulation. However, the underlying background responses need to be taken into account for understanding "true" positive signals. The only raw data for this was shown in the form of a heatmap where no proper ordering was given to ensure that readers can easily interpret the expression of these markers following stimulation relative to no stimulation. Thus, it is difficult to reliably interpret any real differences reported without this. Finally, the authors report differences in either cluster abundance or cluster-specific cytokine/ transcription factor expression in Tfh cell subsets when comparing children vs adults, and between the two malaria antigens. The comparisons of cytokine/transcription factor between groups will be more clearly highlighted by appropriately combining groupings rather than keeping them separate as in Figures 6 and 7.

Thank you for sharing these references. Similar to SPADE clustering and ViSNE dimensionality algorithms used in Orecchioni et al, we used all the extracellular markers from our panel in our FlowSOM algorithm with consensus meta-clustering which includes both the chemokine receptors and activation markers even though they are presented separately in our manuscript across the figure 3 and 4. This was explained in the methods section (lines 573 - 587). We then chose the UMAP algorithm as visual dimensionality reduction of the meta-clusters generated by FlowSOM-consensus meta-clustering as explained under the “OMIQ analysis” subpart of our methods (lines 588- 604). Therefore, we believe we have conducted the analysis as this reviewer suggests even if we chose to show the figures that were informative to our story. The heatmap of the results brings the possibility to see which combination of markers respond or not to the different conditions and between groups, all the raw data are present from the supplemental figures 10 to 13 showing, using bar plots, the differences expressed in the heatmaps. We believe it strengthens our interpretation of the results.

Regarding the transcription factor and cytokine background, we added supplemental figures 14, 15 and 16 where we used manual gating to select Bcl6, cMAF, IFNg, IL21 or IL4 positive cells directly from total CXCR5^{pos}CD4^{pos}CD45RA^{neg}CD25^{neg} TfH cells based on the FMO or negative control, and we overlaid the positive cells on the UMAP of all the CXCR5^{pos}CD4^{pos}CD45RA^{neg}CD25^{neg} meta-clusters. Moreover, all the dot plots (with their statistics) used for the heatmap figure 6 and 7 can be found in the supplemental figures 10, 11, 12 and 13. These supplemental figures address the concerns above by showing the difference of signals between unstimulated and stimulated conditions.

Reviewer #3 (Public Review):

Summary:

> *The goal of this study was to carry out an in-depth granular and unbiased phenotyping of peripheral blood circulating Tfh specific to two malaria vaccine candidates, PfSEA-1A and PfGARP, and correlate these with age (children vs adults) and protection from malaria (antibody titers against Plasmodium antigens.). The authors further attempted to identify any specific differences in the Tfh responses to these two distinct malaria antigens.*
>
> *Strengths:*
>
> *The authors had access to peripheral blood samples from children and adults living in a malaria-endemic region of Kenya. The authors studied these samples using in vitro restimulation in the presence of specific malaria antigens. The authors generated a very rich data set from these valuable samples using cutting-edge spectral flow cytometry and a 21-plex panel that included a variety of surface markers, cytokines, and transcription factors.*
>
> *Weaknesses:*
>
> *- Quantifying antigen-specific T cells by flow cytometry requires the use of either 1- tetramers or 2- in vitro restimulation with specific antigens followed by identification of TCR-activated cells based on de-novo expression of activation markers (e.g. intracellular cytokine staining and/or surface marker staining). Although authors use an in vitro restimulation strategy, they do not focus their study on cells de-novo expressing activation markers as a result of restimulation; therefore, their study is not really on antigen-specific cTfh. Moreover, the authors report no changes in the expression of activation markers commonly used to identify antigen-specific T cells upon in vitro restimulation (including IFNg and CD40L); therefore, it is not clear if their in vitro restimulation with malaria antigens actually worked.*

We understand the reviewer’s point of view and apologies for any confusion. IFNg was expressed but not statistically different between groups. Indeed, looking at the CD8 T cells and using manual gating, we were able to show that IFNg was increased but not statistically significant upon stimulation from $CD4^{pos}CXCR5^{pos}$ cells (supplemental figure 15, panel C), confirming our primary observation using clustering analysis. These results showed that our malaria antigen induced IFNg response in some participants, but not all of them, revealing heterogeneity in this response among individuals within the same group.

Regarding CD40L, in the supplemental figure 7, we can see that some of our meta-clusters expressed more CD40L upon stimulation, but again without leading to statistical differences between groups. Combined with the increased expression of other cytokines and transcription factors, we showed that our stimulation did indeed work. However, because of the high variation within groups, there were no statistical differences across our groups. Because CD40L is not the only marker showing specific T cell activation, and not all T cells respond using this marker alone, a more comprehensive multimarker AIM panel might have highlighted differences between groups. We recognized the limitations of our study and believe that future study will benefit from more activation markers commonly used to identify antigone-specific T cells such as CD69, OX40, 4-1BB (AIM panel), among other markers.

> *- CXCR5+CD4+ memory T cells have been shown to present multi-potency and plasticity, capable of differentiating to non-Tfh subsets upon re-challenge. Although authors included in their flow panel a good number of markers commonly used in combination to identify Tfh (CXCR5, PD-1, ICOS, Bcl-6, IL-21), they only used one single marker (CXCR5) as their basis to define Tfh, thus providing a weak definition for Tfh cells and follow up downstream analysis.*

Sorry for the confusion, even though the subsampled on the $CD4^{pos}CXCR5^{pos}$ $CD25^{neg}$ cells to run our FlowSOM, we showed the different levels of expression across meta-clusters (figure 4 panels A and B) of PD1 (Tfh being PD1 positive cells) and ICOS (indicating the activation stage of the Tfh, "T Follicular Helper Cells" Methods and Protocols book from Springer 2015). We also included an overlay of the manually gated double positive Bcl6-cMAF cells on the $CXCR5^{pos}CD45RA^{neg}CD25^{neg}$ CD4 T cell UMAP plot to show that most of them express Bcl6 (supplemental figure 14). Interestingly, the manually gated IL21 positive cells were less abundant, particularly for children (supplemental figure 15). Because we were not able to include all the markers that are now used to define Tfh cells, we referred to our cell subsets as "TFH-like". This is an acknowledged limitation of our study. Due to the limited blood volume obtained from children and cost of running multiplex flow cytometry assays, our results showing antigen-specific heterogeneity of Tfh subset will have to be validated in future studies that include these additional defining markers.

> *- Previous works have used FACS-sorting and in vitro assays for cytokine production and B cell help to study the functional capacity of different cTfh subsets in blood from Plasmodium-infected individuals. In this study, authors do not carry out any such assays to isolate and evaluate the functional capacity of the different Tfh subsets identified. Thus, all the suggestions for the role that these different cTfh subsets may have in vivo in the context of malaria remain highly hypothetical.*

Unfortunately, low blood volumes obtained from children prevented us from running *in vitro* functional assays and the study design did not allow us to correlate them with protection. However, since the function of identified Tfh subsets from malaria-exposed individuals has been evaluated using Pf lysates in other studies, we referenced them when interpreting the differences we reported in Tfh subset recognition between malaria antigens. If either of these antigens move forward into vaccine trials, then evaluating their function would be important.

> *- The authors have not included malaria unexposed control groups in their study, and experimental groups are relatively small (n=13).*

This study design did not include the recruitment of malaria naive negative controls as its goal was to assess malaria antigen-specific responses comparing the quality and abundance between malaria-exposed children to adults to these potential new vaccine targets *Pf*SEA-1A and *Pf*GARP. We did however test 3 malaria-naive adults and found no non-specific activation after stimulation with these two malaria antigens. Since this was done as part of our assay optimization, we did not feel the need to show these negative findings.

And even with our small sample size, we demonstrated significant age-associated differences in malaria antigen-specific responses from cT_{FH}-like subsets.

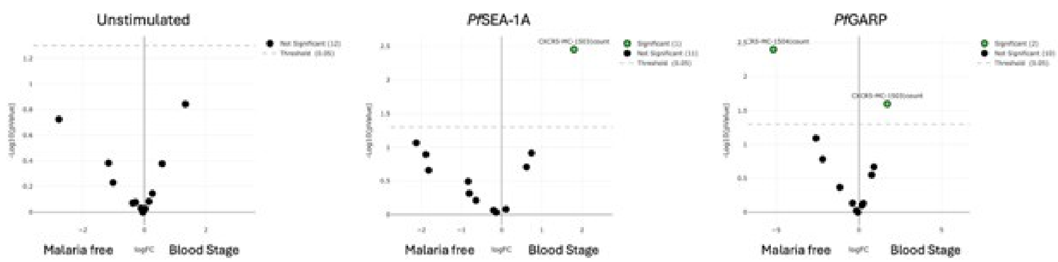
> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *Minor points are:*
>
> *(1) Line 88, cTfh cells are not only from GC-Tfh, they have GC-independent origin (He et al, PMID: 24138884).*

The following sentence was added line 88 "Interestingly, cT_{FH} cells can also come from peripheral cT_{FH} precursor $CCR7^{low}PD1^{high}CXCR5^{pos}$ cells; thus, they also have a GC-independent origin (He, Cell, 2013 PMID: 24138884).

> *(2) I believe all participants were free of blood-stage infection upon enrolment. But can authors clearly state this information between lines 151-159?*

We mentioned in the methods, line 495-496 "Participants were eligible if they were healthy and not experiencing any symptoms of malaria at the time venous blood was collected". However, using qPCR we found 5 children with malaria blood stage. As shown in Author response image 2, comparing malaria free to blood-stage children, no differences were observed without any stimulation. However, MC03 is more abundant upon malaria antigen stimulation in the blood-stage group whereas MC04 is more abundant in the malaria free group upon *Pf*GARP stimulation only confirming that our stimulation worked.

Author response image 2.



> ***Reviewer #3 (Recommendations For The Authors):***
>
> *(1) The strategy for gating on antigen-specific cTfh cells needs to be revised. The correct approach would be to gate on those cells that respond by de-novo expression of activation markers upon antigen restimulation (also termed activation-induced markers. e.g. CD69, CD40L, CXCL13 and IL-21, Niessl 2020; CD69, CD40L, CD137 and OX40, Lemieux 2023; CD137 and OX40, Grifoni 2020). As it stands, the study is not really on antigen-specific T cells, but rather on the overall CD4 T cell compartment plus or minus antigenic stimulation.*

We recognized the limitation in our flow panel design which prevents us from performing this gating. We originally based our panel design on the "T follicular helper cells methods and protocols" book (Springer 2015) which used CD45RA, CD25, CXCR5, CCR6, CXCR3, CCR7, ICOS and PD1 to define cT_{FH}. We had already optimized our 21-color panel, purchased reagents and started to run our experiments by the time these publications modified how to define TFH cells Niessl, Lemieux and Grifoni's publication. Indeed we optimized and performed our assay from November 2019 to March 2020, finishing to run the samples during the first quarantine. Because of the urgent needs of research on SARS-CoV-2 that we were involved with from this time and moving forward, the analysis of our TFH work got highly postponed. Moreover, 2020 is also the year where many TFH papers came out with better ways to define cT_{FH} and responses to antigen stimulations. In our future studies, our panel will include AIM.

> *(2) It is not clear if the antigenic stimulation actually worked. Does the proportion of IFNg+ or IL-4+ or IL-21+ or CD40L+ or CD25+ CD4 or CD8 T cells increase following in vitro antigen restimulation?*

Yes, using manual gating, we are able to show an increase of IL4 (supplemental figure 16 panel B and C), and IL21 (supplemental figure 15 panel J and K) production in both children and adults. However, we did not observe significant production of IFNg (supplemental figure 15, panel C) and changes in CD40L expression (supplemental figure 7) after malaria antigen stimulation, however, our positive control SEB worked. So, yes our stimulation assay worked but these 2 malaria antigens did not significantly induce these cytokines. This could be that they are too low to detect in every participant since they are single antigens and not whole parasite lysates, as other studies have used. It could also be that these antigens don't stimulate CD40L or IFNg in all our participants. We brought up this limitation as follow in the discussion, line 473: "Although the heterogeneity in the response of CD40L and IFNγ suggests that our tested malaria antigens did not induce significant differences in the expression of these markers in all our participants, our panel did not include other activated induced markers, such as OX40, 4-1BB, and CD69".

> *(3) It is not clear what is the proportion of cTfh over the total CD4 T cell compartment among the different groups. Does this vary among different groups? It would be valuable to display this as an old-fashioned combination of contour plots with outliers for illustrating flow cytometry and bar graphs for the cumulative data.*

The proportion of $CD3^{pos}CD4^{pos}CD25^{neg}CXCR5^{pos}$ cTfh cells did not differ within the total number of CD4 T cells between groups (figure 2).

> *(4) The gating strategy could be refined and become more robust if adding additional markers in combination with CXCR5 for identifying cTfh (e.g. CXCR5+Bcl6+).*

Thank you for this suggestion. An overlay of Bcl6 expression can be found in supplemental figure 14 where we confirm that our CXCR5+ cT_{FH}-like subsets express cMAF and Bcl6.

> *(5) The protocols for intracellular and intranuclear staining seem to be incomplete in Materials and Methods. In particular, cell permeabilization strategies seem to be missing.*

Our apologies for this oversight, we added the following sentences in the methods line 545: "Cells were fixed and permeabilized for 45 mins using the transcription factor buffer set (BD Pharmingen) followed by a wash with the perm-wash buffer. Intracellular staining was performed at 4 °C for 45 more mins followed by two washes using the kit's perm-wash buffer".

> *(6) In Materials and Methods, the authors mention they have used fluorescence minus one control to set their gating strategy. It would be valuable to show these, either on the main body or as part of supplementary figures.*

We added the cytoplots of the FMOs and/or negative controls as appropriate in the supplemental figures 14 (cMAF and Bcl6), 15 (IFNg and IL21) and 16 (IL4 and IL21).

> *(7) Line 194 and Figure 3, it is not clear the criteria that the authors used for down-sampling events before FlowSOM analysis. Was this random? Was this done with unstimulated or stimulated samples?*

We chose to down-sample on $CD3posCD4^{pos}CD25^{neg}CD45RA^{neg}$ and $CXCR5^{pos}$ cells prior to our FlowSOM to allow more cluster analysis to focus only on the differences among those cells. The down-sampling used 1,000 $CD3posCD4^{pos}CD25^{neg}$ $CD45RA^{neg}CXCR5^{pos}$ cells from each fcs file (unstimulated and stimulated samples). If the fcs file had more than 1,000 $CXCR5^{pos}$ cells, the down-sampling was done randomly by the OMIQ platform algorithm to select only 1,000 $CXCR5^{pos}$ cells within this specific fcs file. The latest sentence was added to the methods line 593.

> *(8) Lanes 201, 202, As it stands, the take of the authors on the role of different cTfh subsets during infection remains highly speculative. Are these differences in cTfh phenotypes actually reflected in their in vitro capacity to provide B cell help (e.g. as in the Obeng-Adjei 2015 paper) or to produce IL-21, express co-stimulatory molecules, or any other characteristic that would allow them to better infer their functional roles during infection? Any additional in vitro analysis of the functional capacity of isolated cTfh subsets identified in this research would greatly increase its value.*

We agree with the reviewer that this sentence is speculative, and we rephrase it as follow: "First, we found different CXCR5 expression levels between meta-clusters (Figure 3b); CXCR5 is essential for cT_{FH} cells to migrate to the lymph nodes and interact with B-cells". We would have liked to perform *in vitro* functional assays. However, as explained above, we did not have sufficient cells collected from children to do so.

> *(9) It is not clear why authors omitted IL-17 and did not use IFNg and IL-4 to refine their definition of Th1, Th2 and Th17 cTfh.*

We would have liked to include IL-17, however we were constrained by only having access to a 4 lasers cytometer at the time we ran our assay. In light of needing to prioritize markers, when we were designing our flow panel, cTfh1 were shown to be preferentially activated during episodes of acute febrile malaria children (Obeng-Adjei). Therefore, we chose to focus on IFNg and IL4 to differentiate Tfh1 from Tfh2, in addition to other markers as surrogate of functional potential. We did not use IFNg and IL4 to refine our definition of Tfh1, Tfh2 and Tfh17 as recent publications have shown that IL4 is not only expressed in Tfh2 but also in the other Tfh subsets, at lower intensity (Gowthaman among others). Therefore IFNg and IL4 by themselves were not sufficient to properly define the different Tfh subsets. In future studies, we plan to include transcription factor profiles (T-bet, BATF, GATA3) to further refine definitions of Tfh subsets.

> *(10) Lines, 226, 228, based on the combination of markers that the MC03 subset expresses, it is tempting to think that this is the only "truly" committed Tfh subset from the entire analysis. Please, discuss.*

If the reviewer is referring to changes in marker expression levels that indicate they have not reached a level of differentiation that would make them reliable (ie "true) Tfh cells, we agree that this is an important question now that we have technology that can measure and analyse so many phenotypic markers at once. This brings forward the need for the scientific method - to replicate study findings to determine whether they are consistent given the same study design and experimental conditions.

> *(11) Lines 243 244, Again, is this reflected in functional capacity?*

The study described in this manuscript did not include functional assays. However, this did not change the key finding that different malaria antigens behaved differently, demonstrating heterogeneity in Tfh recognition of malaria antigens. Regarding CD40L expression, we did not observe differences between groups, however some individuals had an increase of their CD40L (supplemental figure 7). It is possible that some individuals had responded through other activated induced markers (CD69, ICOS, OX40, 4-1BB among others) and that our stimulation condition was not long enough to assess CD40L expression upon malaria antigen stimulation. This limitation has been addressed by editing the line 243-244 as follows: "we were unable to find statistical differences in the CD40L expression between groups as only few individuals responded through it (supplemental figure 7)."

> *(12) Lines 243, 244, Are these cTfh subsets exclusively detected in malaria-exposed individuals? This is confounded by the lack of a malaria unexposed control group in this study, which would have been highly valuable.*

We agree with the reviewer that having non-naive children would have been valuable as a negative control group. However, this study was conducted in Kenya where all children are suspected to have had at least one malaria infection. We also did not have ethical approval or the means to enroll children in the USA who would not have been exposed to malaria as a negative control group. Since we were also evaluating differences by age group, comparing US adults would not have helped to address this point. Therefore, this remains an open question that might be addressed by another study recruiting children in non-malaria endemic areas.

> *(13) Line 267, as the authors have not gated on T cells de-novo expressing activation markers in response to antigen restimulation, how do they know these are indeed antigen-specific cTfh?*

Omiq analysis accounts for marker expression levels in the resting cells (unstimulated well) for each individual compared to each experimental/stimulated well. The algorithm computationally determines whether that expression level changed without an arbitrary positive threshold, keeping the expression levels as a continuous variable, not dichotomous - which is the power of unbiased cluster analyses. Therefore, we know that these cells are antigen-specific based on the statistical difference in intensity expression between the resting cells and the stimulated ones. Nevertheless, manual gating to show “de-novo” responding cells, produced the same results as assessing the MFI of each meta-cluster (supplemental figures 14, 15 and 16).

> *(14) Lines, 292-295, it is very surprising that Tfh cells would not produce IL-21 upon restimulation. Have the authors observed upregulation of IL-21 following SEB restimulation?*

Yes, we observed IL21 positive cells upon SEB stimulation (supplemental figure 15, panel J and K). However we found unexpectedly high background levels of IL21, specifically within the adult group (supplemental figure 15, panel K and M) making it challenging to find antigen-specific increases above background. Interestingly, an increase in IL21 using manual gating was observed upon *Pf*SEA-1A or *Pf*GARP stimulation in children (supplemental figure 15, panel J and L).

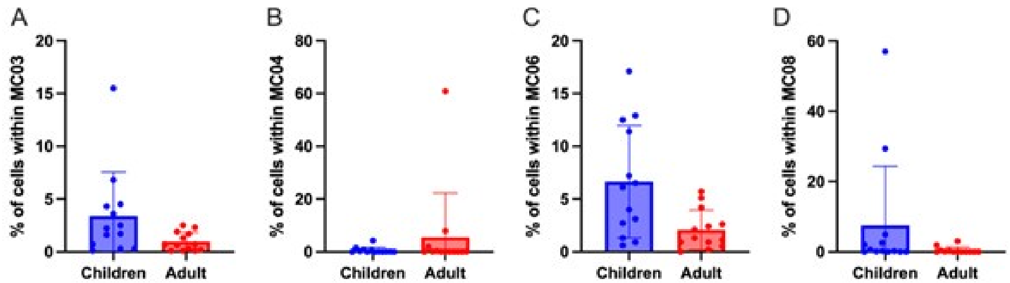
> *(15) In Figures 3 and 4, it is not clear if there are any significant differences in expression of different markers between different cTfh subsets and/or different conditions. Moreover, the lack of differences in response to antigen stimulation seems to suggest that it did not work adequately.*

We intentionally chose 6-hours stimulation to better assess changes in cytokines which we did. However, because it is a short stimulation, we did not expect dramatic changes in the extracellular markers presented in the figure 3 and 4. A longer stimulation, such as 24h, will highlight properly these changes.

> *(16) Figure 5b would benefit from bar graphs.*

Please find below the bar-graphs for the highlighted meta-clusters in figure 5b. We did not include these bar-graphs to our figure 5 as they do not bring new information. They repeat the information already presented through the EdgeR plot.

Author response image 3.



> *(17) Figures 6 and 7 would greatly benefit from showing individual examples of old-fashioned contour with outliers flow plots to illustrate the different cTfh subsets identified in the study.*

The different cT_{FH} subsets can be found with a contour plot with outliers in the supplemental figure 4.

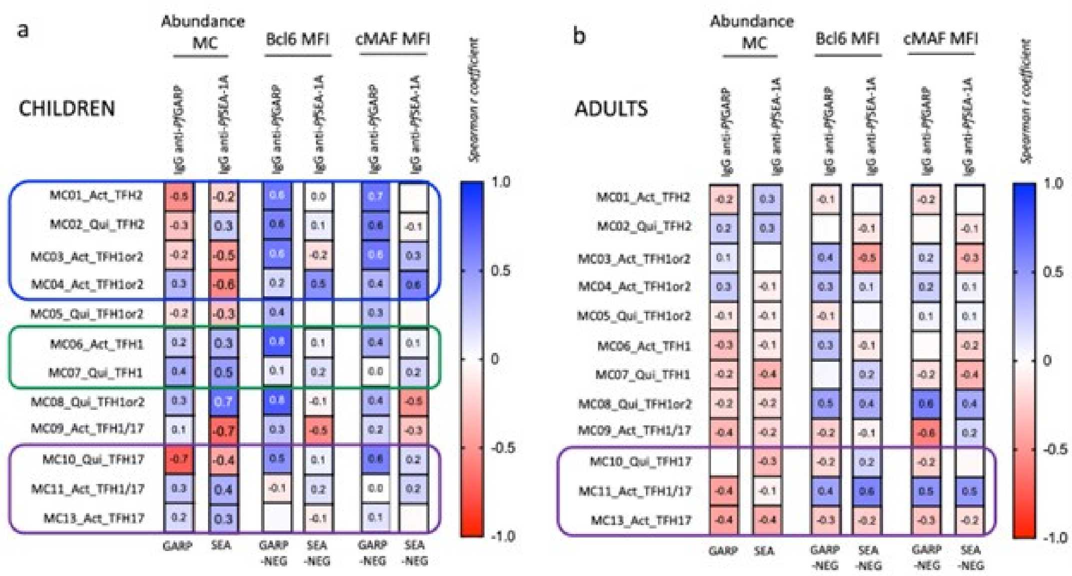
> *(18) Figures 3,4, 6, and 7, the authors exclusively focused on the study of MFI to measure the expression of cytokine and transcription factors among different groups/stimulations. Have the authors observed any differences in the percentage or absolute counts of cytokine+ and/or TF+ between different subsets of cTfh and/or different conditions?*

Yes. We added the supplemental figures 14 (transcription factors) and 15/16 (cytokines) where cytokines and transcription factors were assessed using manual gating. We found that total $CD4^{pos}CXCR5^{pos}$ IL4 was significantly increased upon stimulation in both adults and children while IFNg was not. However, we found significantly higher IFNg on total $CD8^{pos}$ cells showing that the stimulation worked, but the total $CD4^{pos}CXCR5^{pos}$ did not express IFNg. Finally, we observed a trend of higher $IL21^{pos}CD4^{pos}CXCR5^{pos}$ in adults, not significant due to high background whereas IL21 was significantly increased upon stimulation in children. Regarding cMAF and Bcl6, both transcription factors were significantly increased upon stimulation within children only.

> *(19) Figure 8, the definition for high and low PfGARP antibody titers seems rather arbitrary. Are these associations still significant when attempting a regular correlation analysis between Ab values (i.e. Net MFI) and different cTfh subsets?*

Yes, the definition for high and low *Pf*GARP antibody levels is arbitrary but when looking at the antibody data (figure 1b), it was naturally bimodal. Therefore as a sub-analysis, we assess the association between *Pf*GARP antibodies levels and cT_{FH} subsets, see Author response image 4. We checked the correlation between the abundance of the meta-clusters and the level of IgG anti-*Pf*GARP and anti-*Pf*SEA after *Pf*GARP and *Pf*SEA stimulation. We also checked the correlation between the MFI expression of Bcl6 and cMAF after stimulation (*Pf*GARP or *Pf*SEA-1A minus the unstimulated) by the meta-clusters and the level of IgG anti-*Pf*GARP and anti-*Pf*SEA. However, we believe that because of our small sample size, our results are not robust enough and that we risk over-interpreting the data. Therefore, we choose not to include this analysis in the manuscript.

Author response image 4.



> *(20) The comprehensive 21-plex panel that authors used in this study could generate insights on additional immune cells beyond cTfh (e.g. additional CD4 T cell subsets, CD8 T cells, CD19 B cells). It is not clear why the authors limited their analysis to cTfh only.*

The primary goal of the study was to assess the cT_{FH} response to malaria vaccine candidates. However, we were able to assess the IFNg expression for CD8 T cells upon stimulation using the manual gating as indicated in the supplemental figure 15. Without additional markers to more clearly define other CD4 T cell or B cell subsets, we do not believe this dataset would go deep enough into characterizing antigen-specific responses to malaria antigens that would yield new insight.

> *(21) Minor point, the punctuation should be revised throughout the manuscript.*

Punctuation was revised throughout the manuscript by our departmental scientific writer Dr. Trombly, as per reviewer request.

https://doi.org/10.7554/eLife.98462.2.sa0